

BFS Traversal

Step 1: Start at A

Legend

Complete

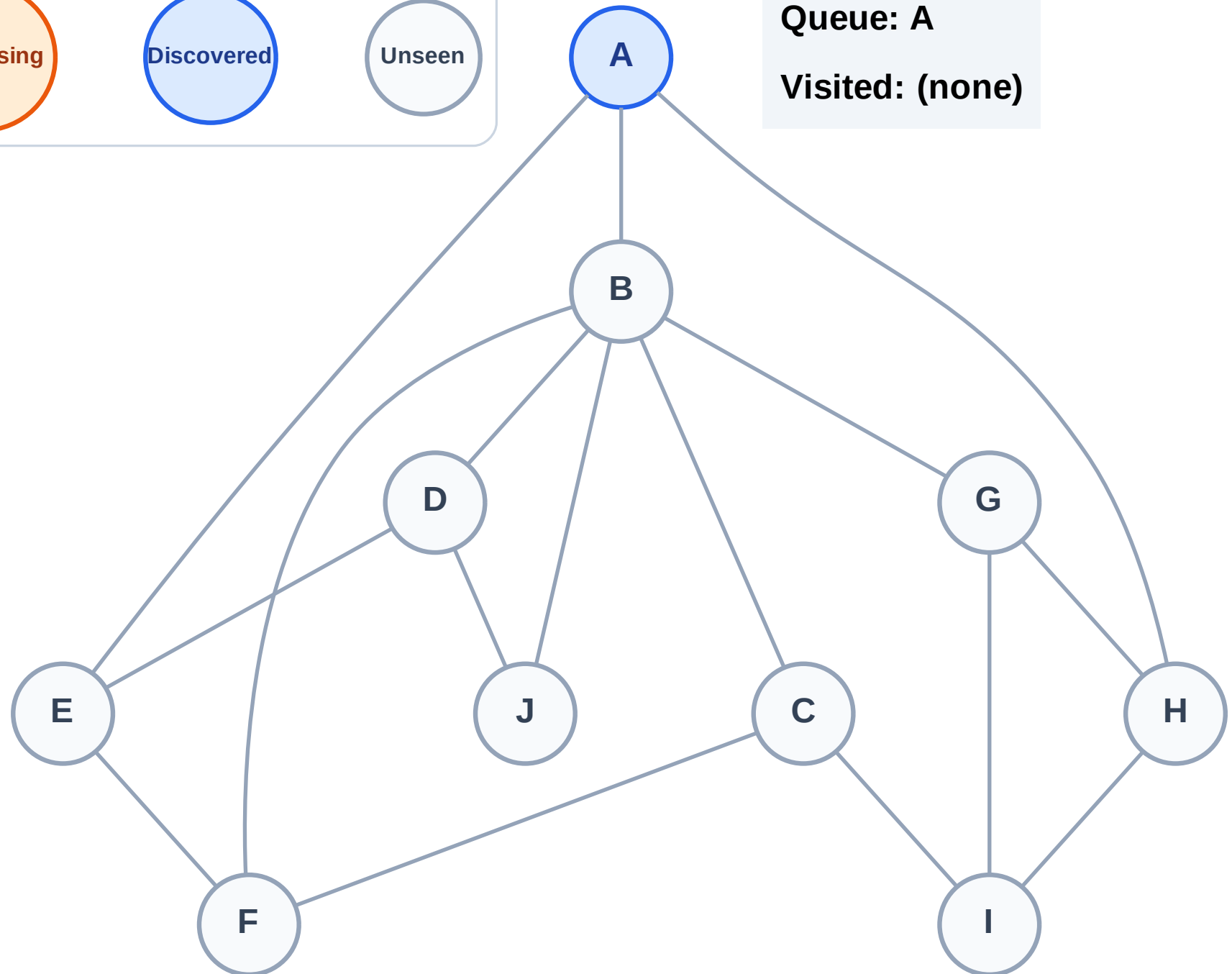
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

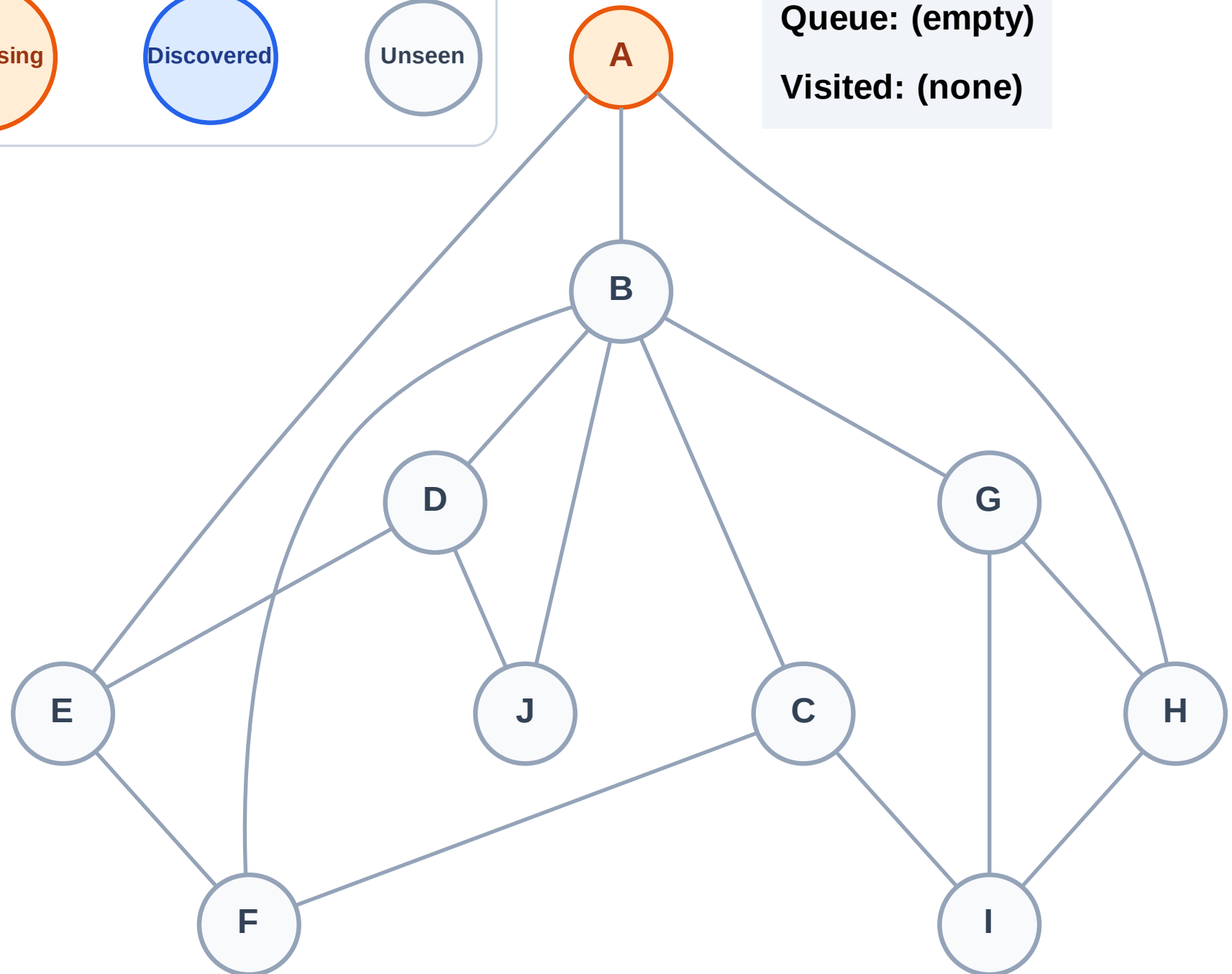
Processing

Discovered

Unseen

Queue: (empty)

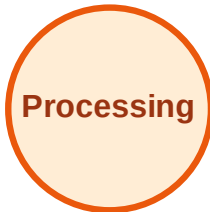
Visited: (none)



BFS Traversal

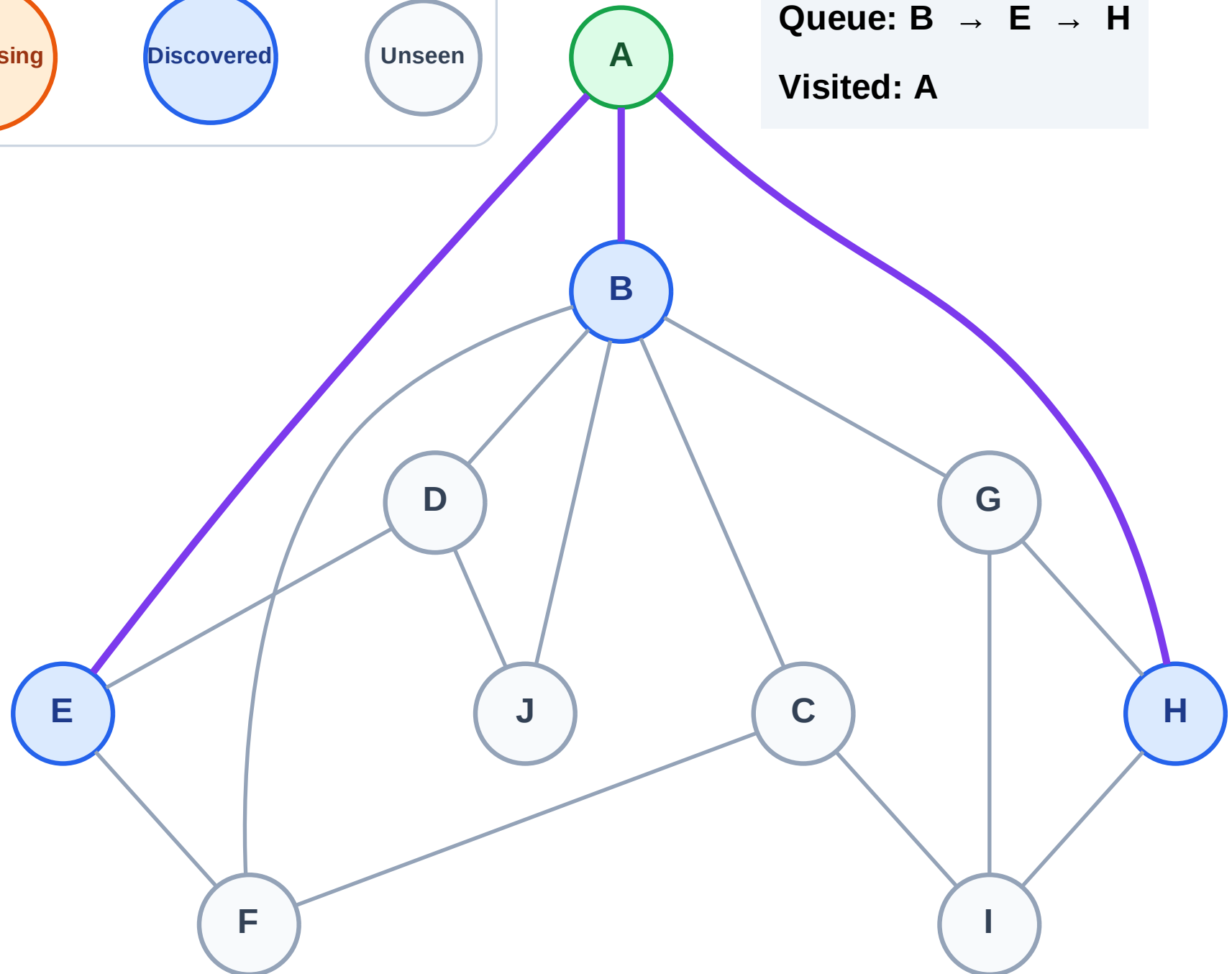
Step 3: Discover B, E, H from A

Legend



Queue: B → E → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

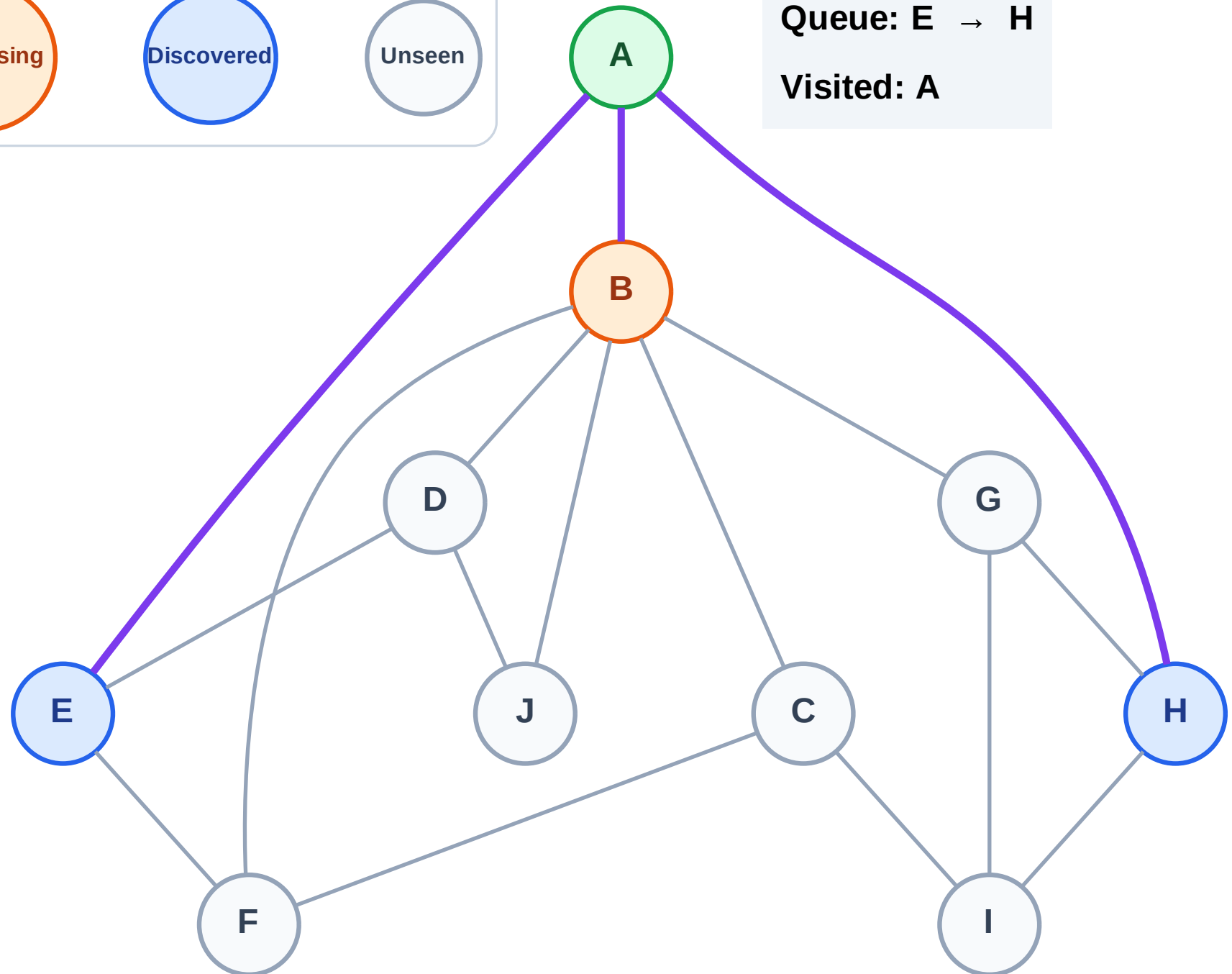
Processing

Discovered

Unseen

Queue: E → H

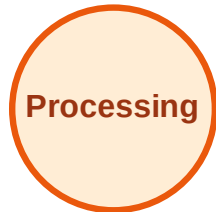
Visited: A



BFS Traversal

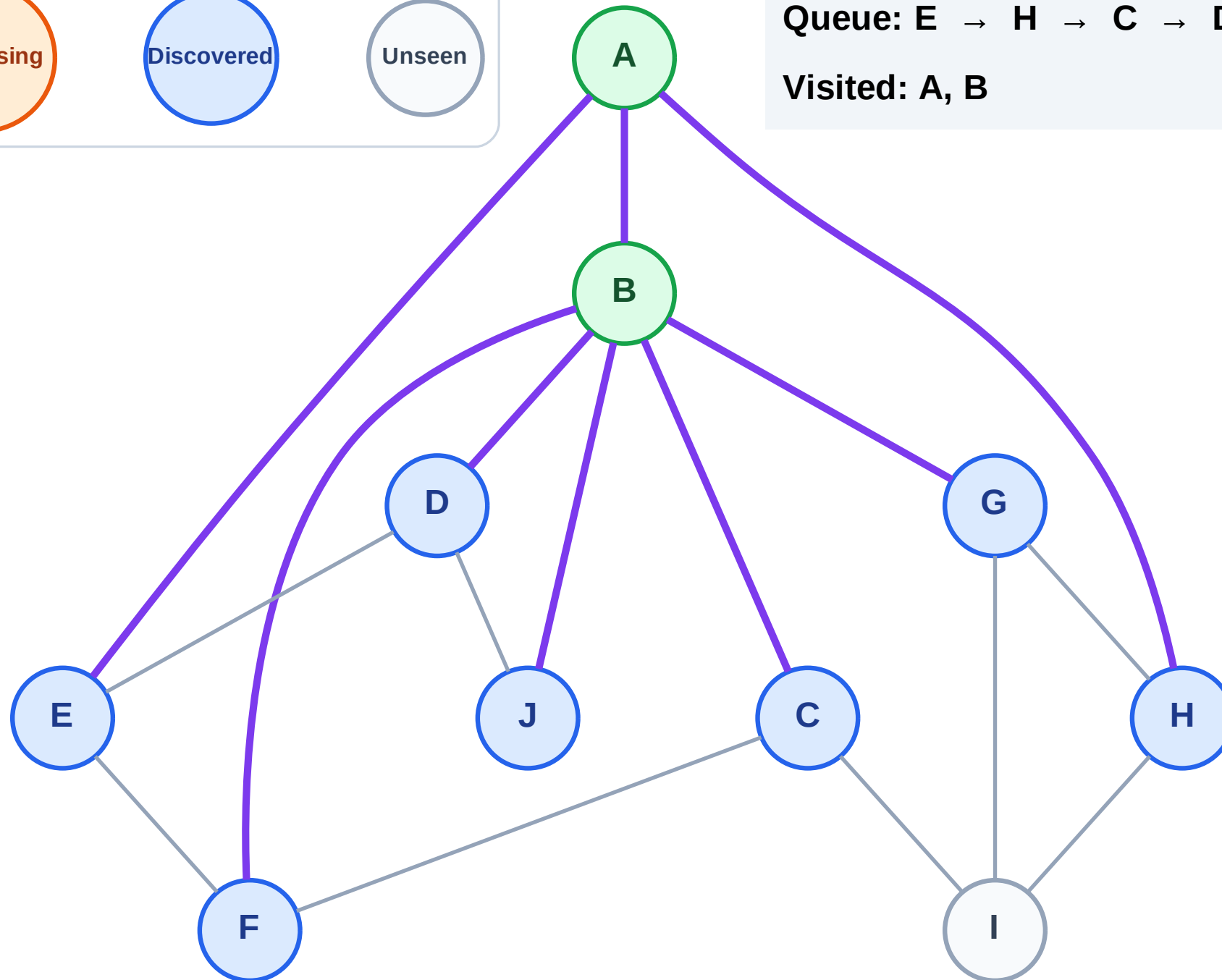
Step 5: Discover C, D, F, G, J from B

Legend



Queue: E → H → C → D → F → G → J

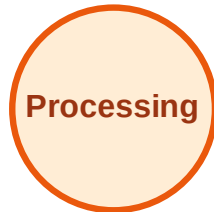
Visited: A, B



BFS Traversal

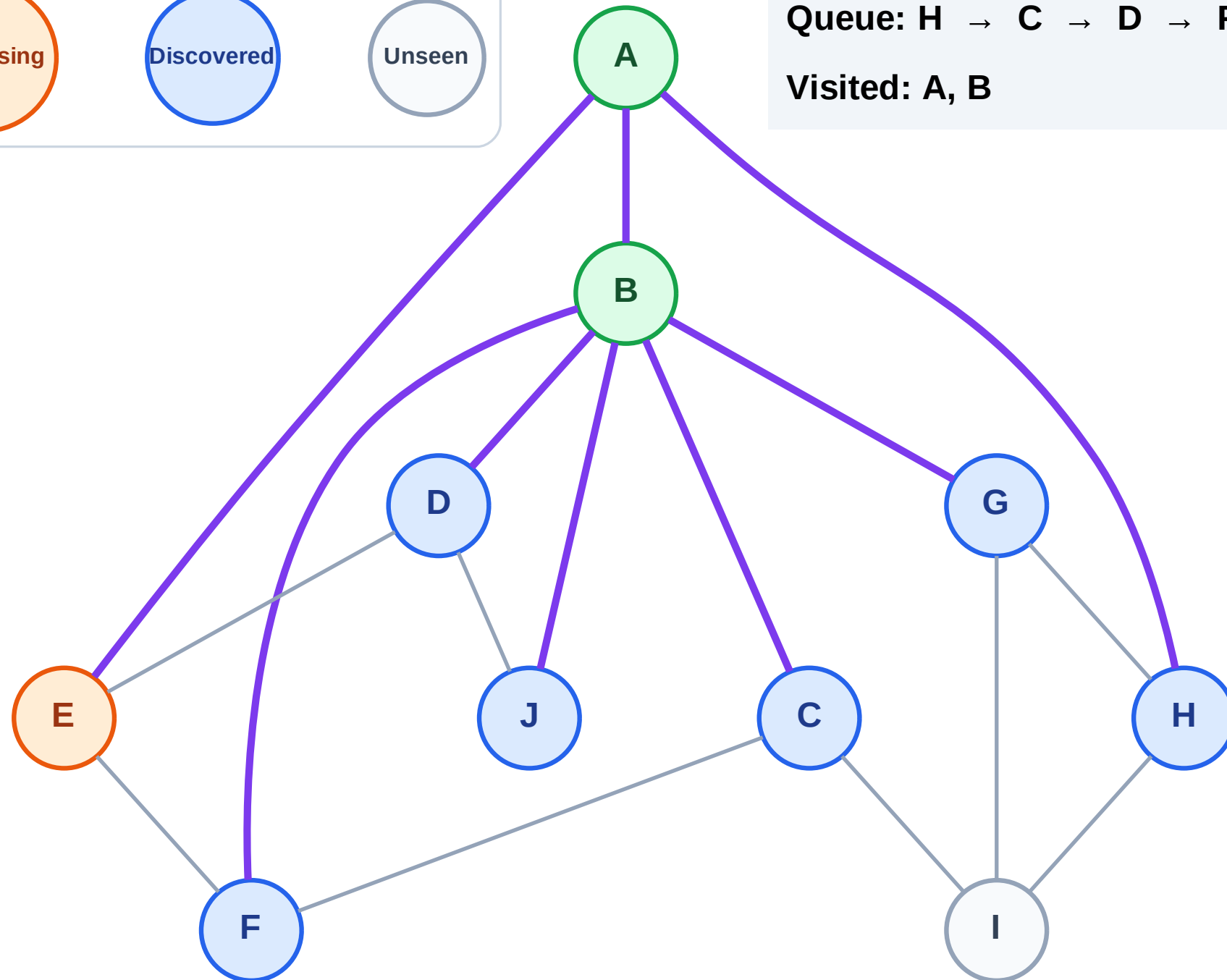
Step 6: Process node E

Legend



Queue: H → C → D → F → G → J

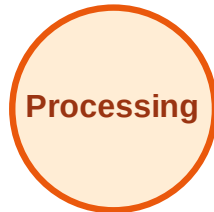
Visited: A, B



BFS Traversal

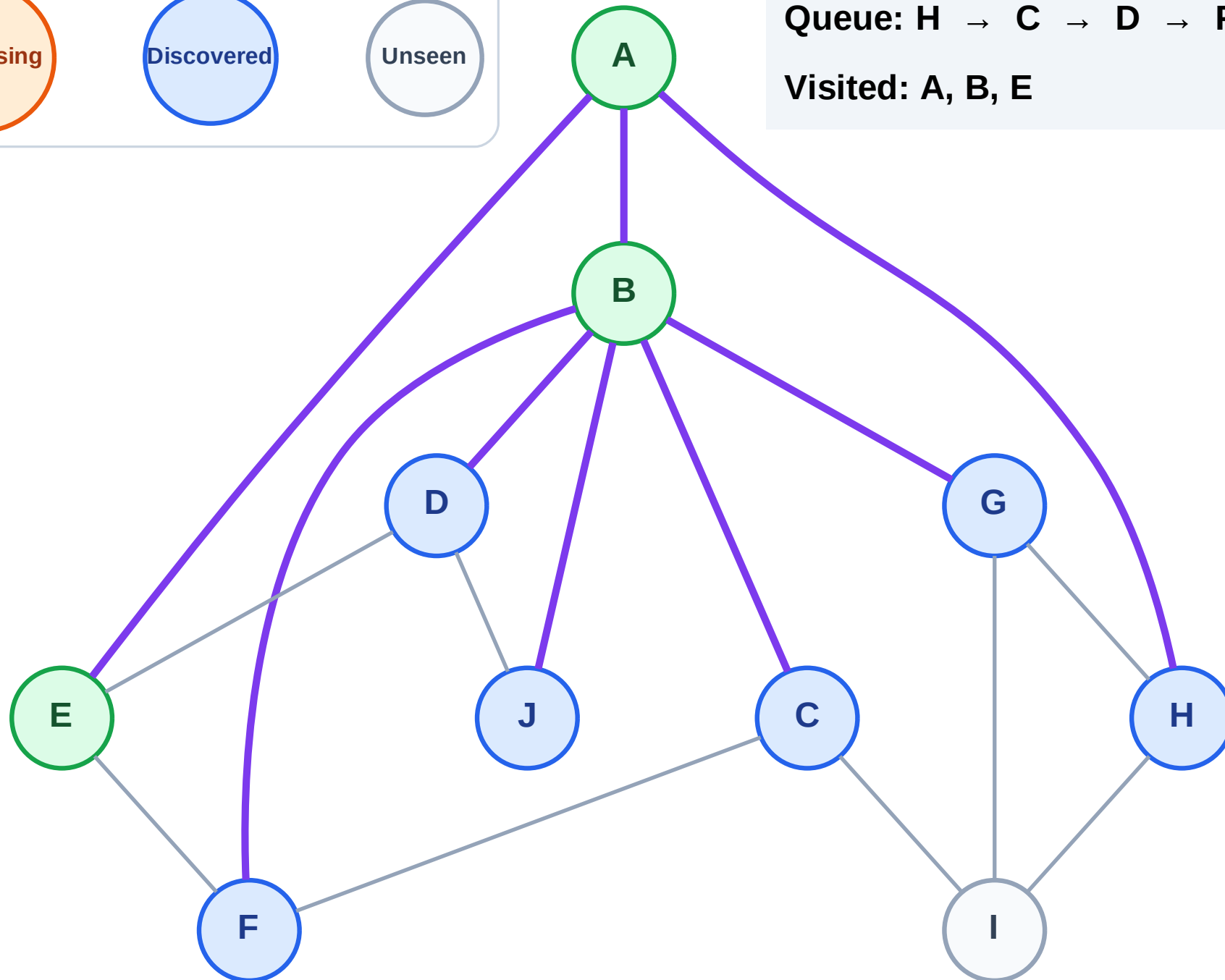
Step 7: No new neighbors from E

Legend



Queue: H → C → D → F → G → J

Visited: A, B, E



BFS Traversal

Step 8: Process node H

Legend

Complete

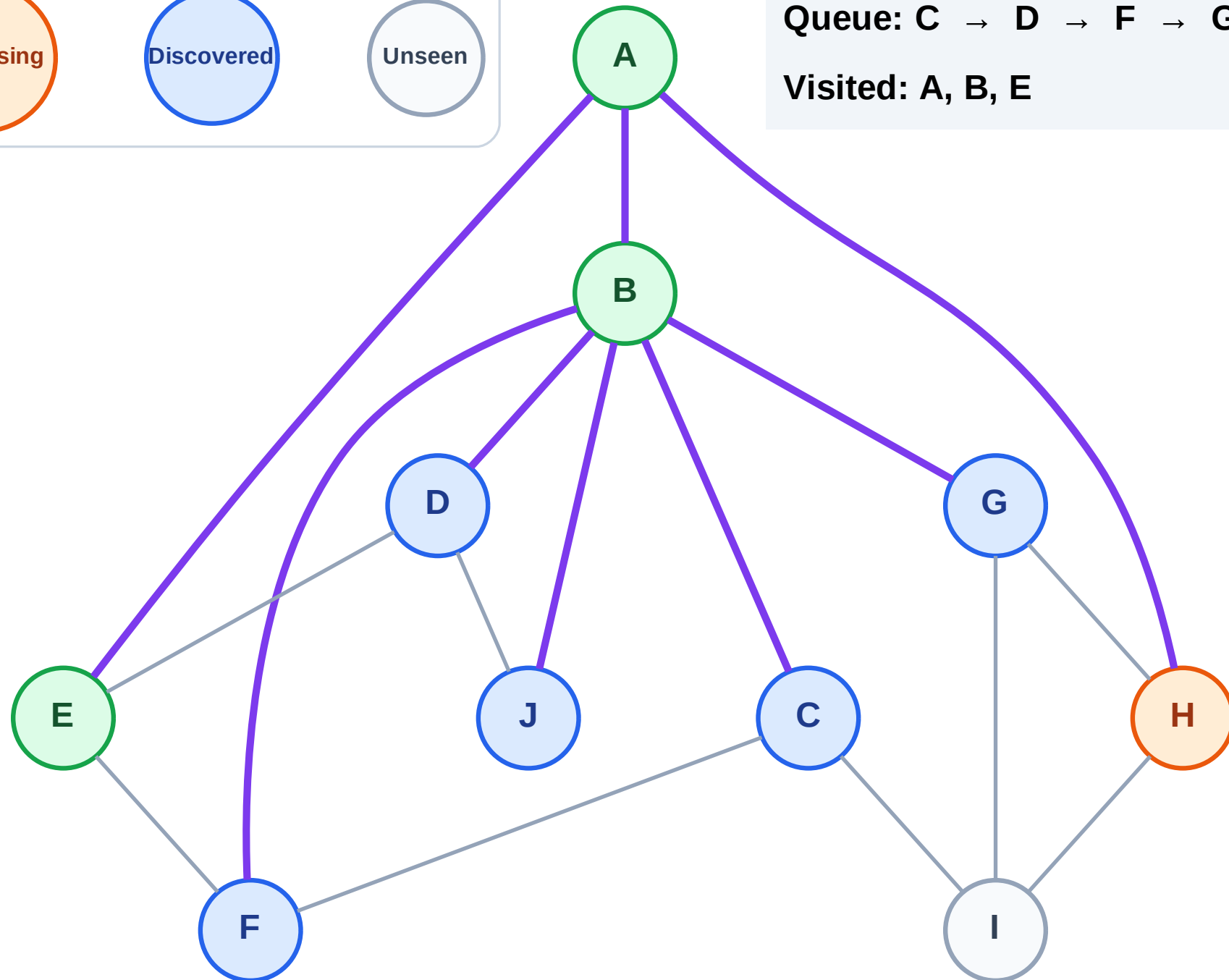
Processing

Discovered

Unseen

Queue: C → D → F → G → J

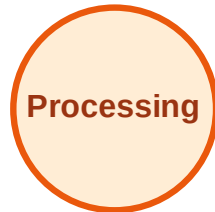
Visited: A, B, E



BFS Traversal

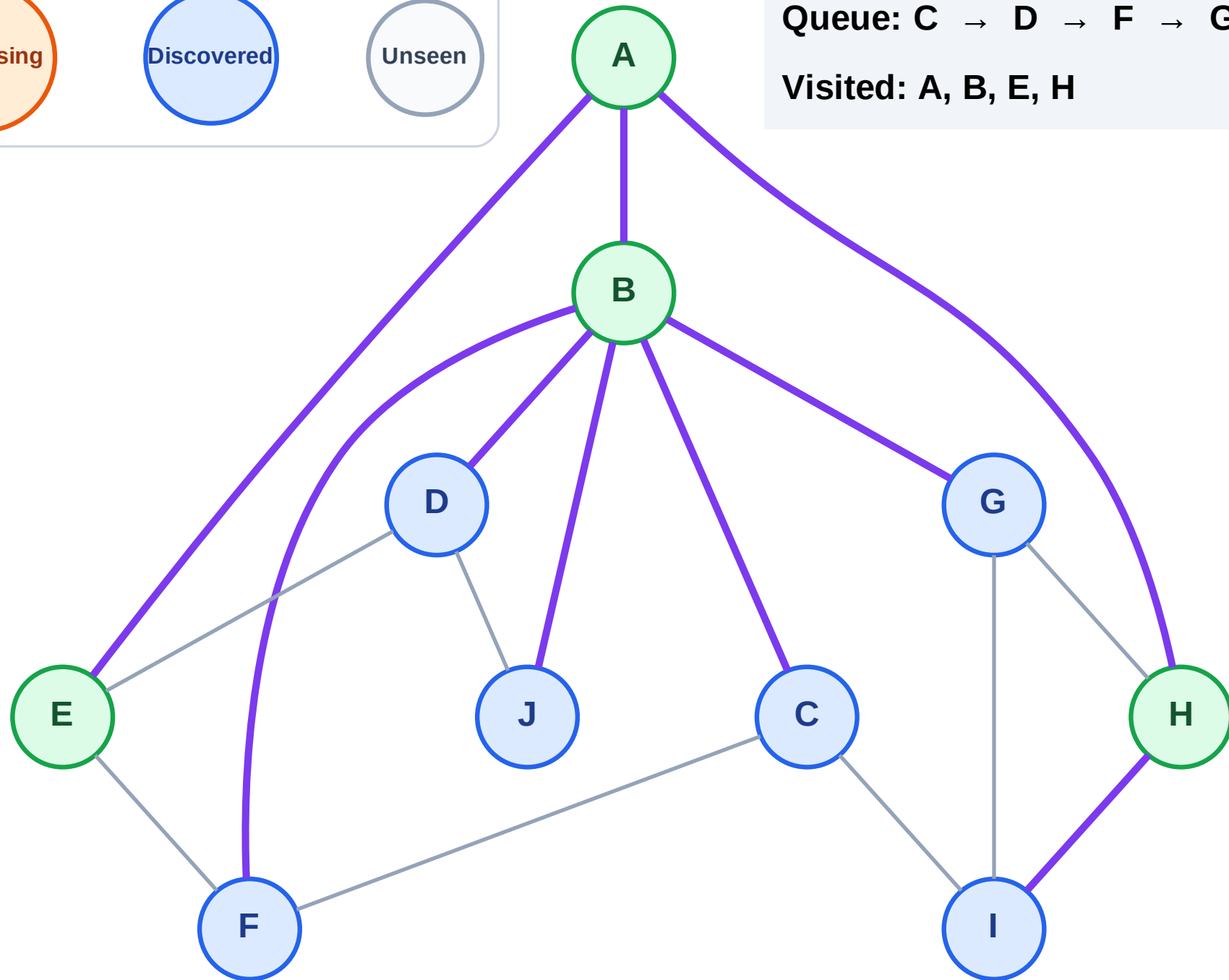
Step 9: Discover I from H

Legend



Queue: C → D → F → G → J → I

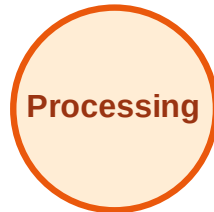
Visited: A, B, E, H



BFS Traversal

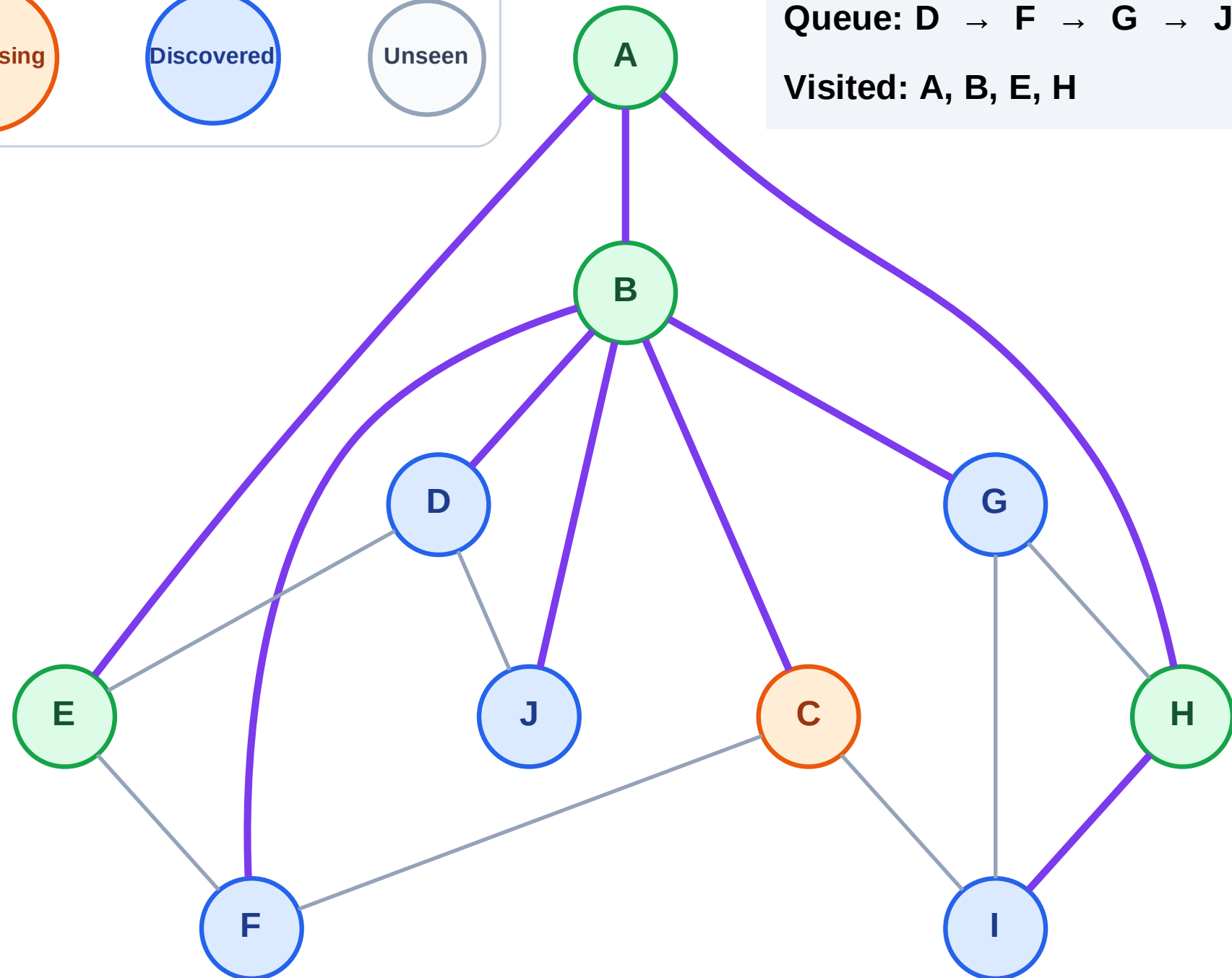
Step 10: Process node C

Legend



Queue: D → F → G → J → I

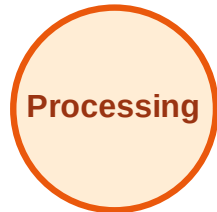
Visited: A, B, E, H



BFS Traversal

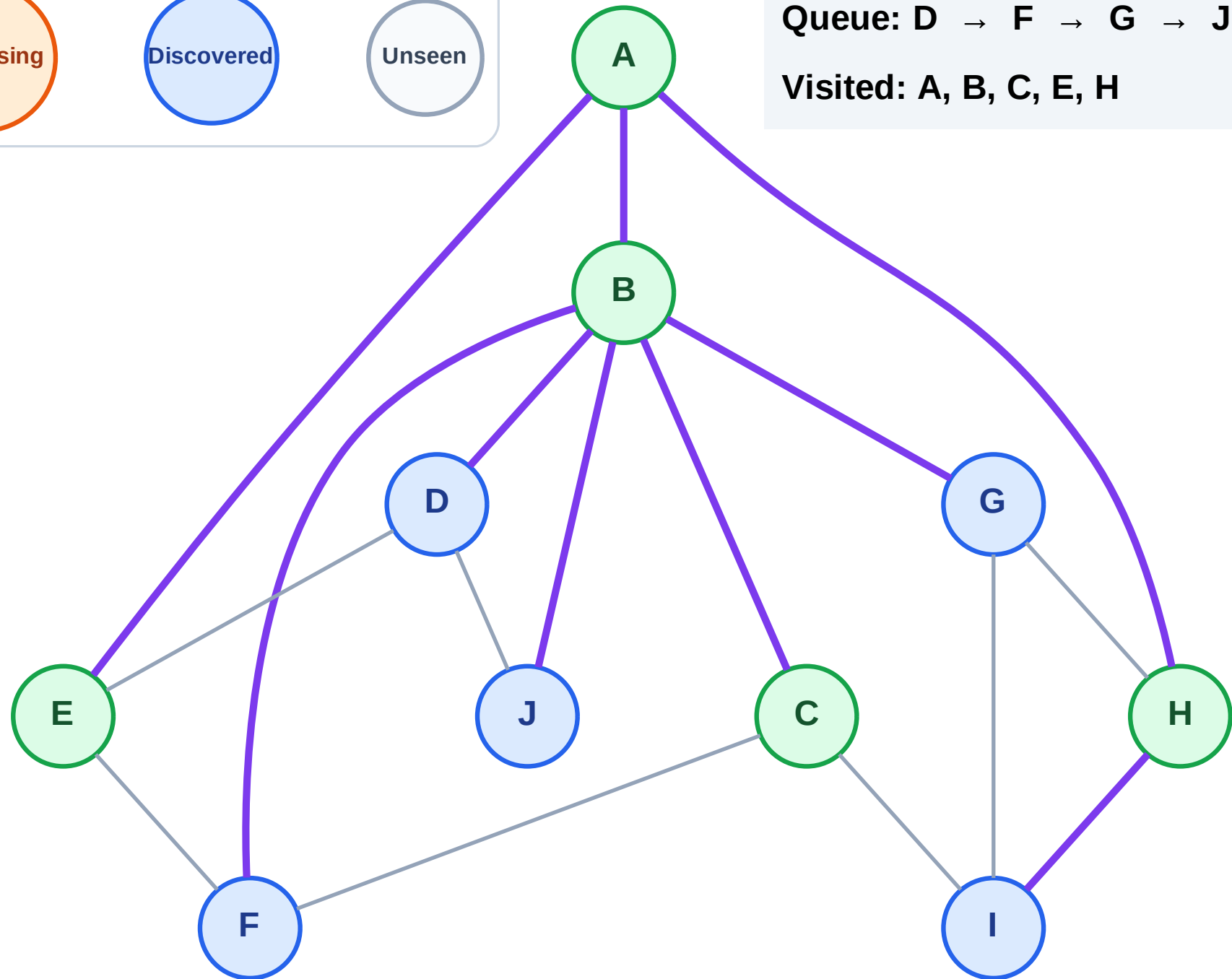
Step 11: No new neighbors from C

Legend



Queue: D → F → G → J → I

Visited: A, B, C, E, H



BFS Traversal

Step 12: Process node D

Legend

Complete

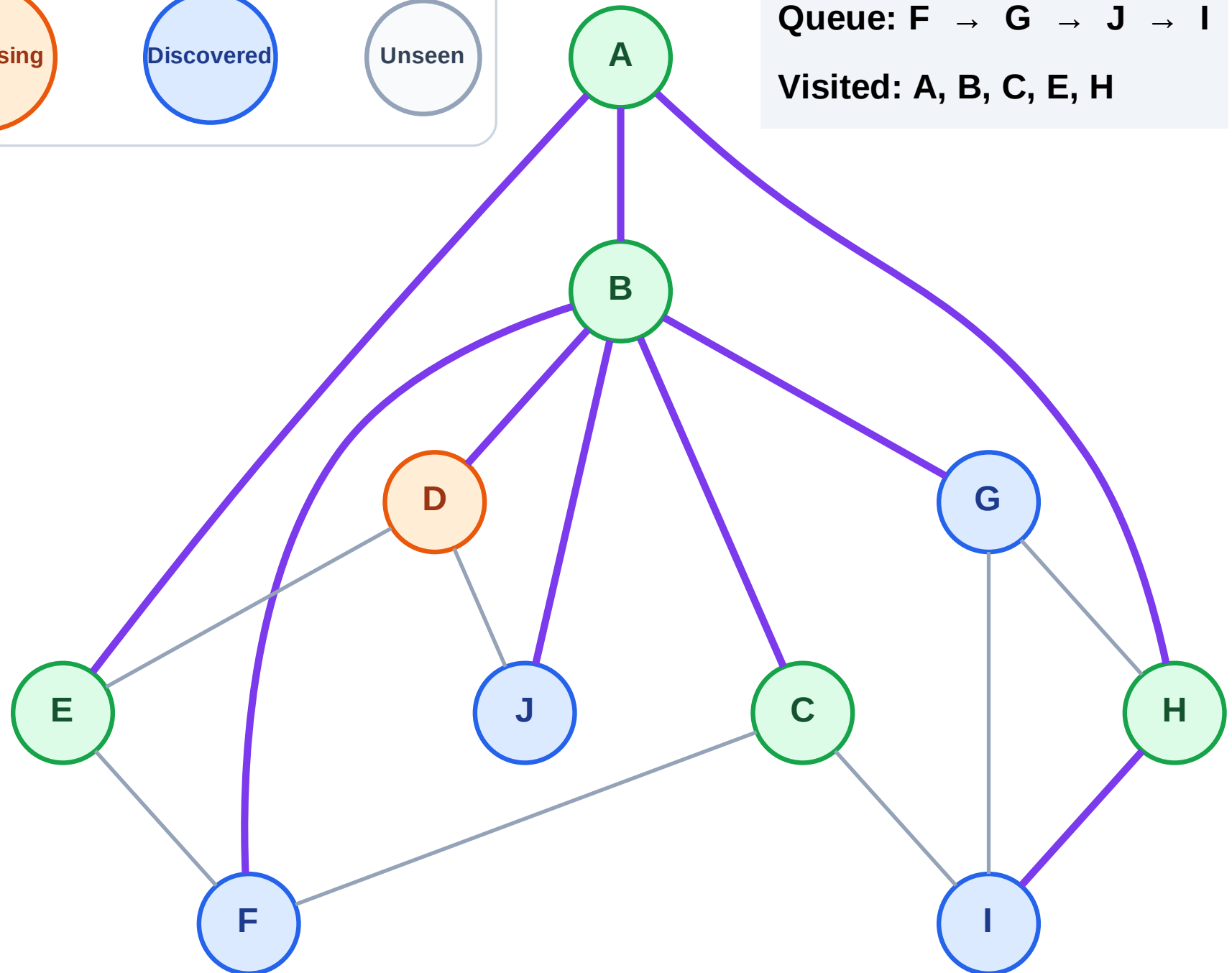
Processing

Discovered

Unseen

Queue: F → G → J → I

Visited: A, B, C, E, H



BFS Traversal

Step 13: No new neighbors from D

Legend

Complete

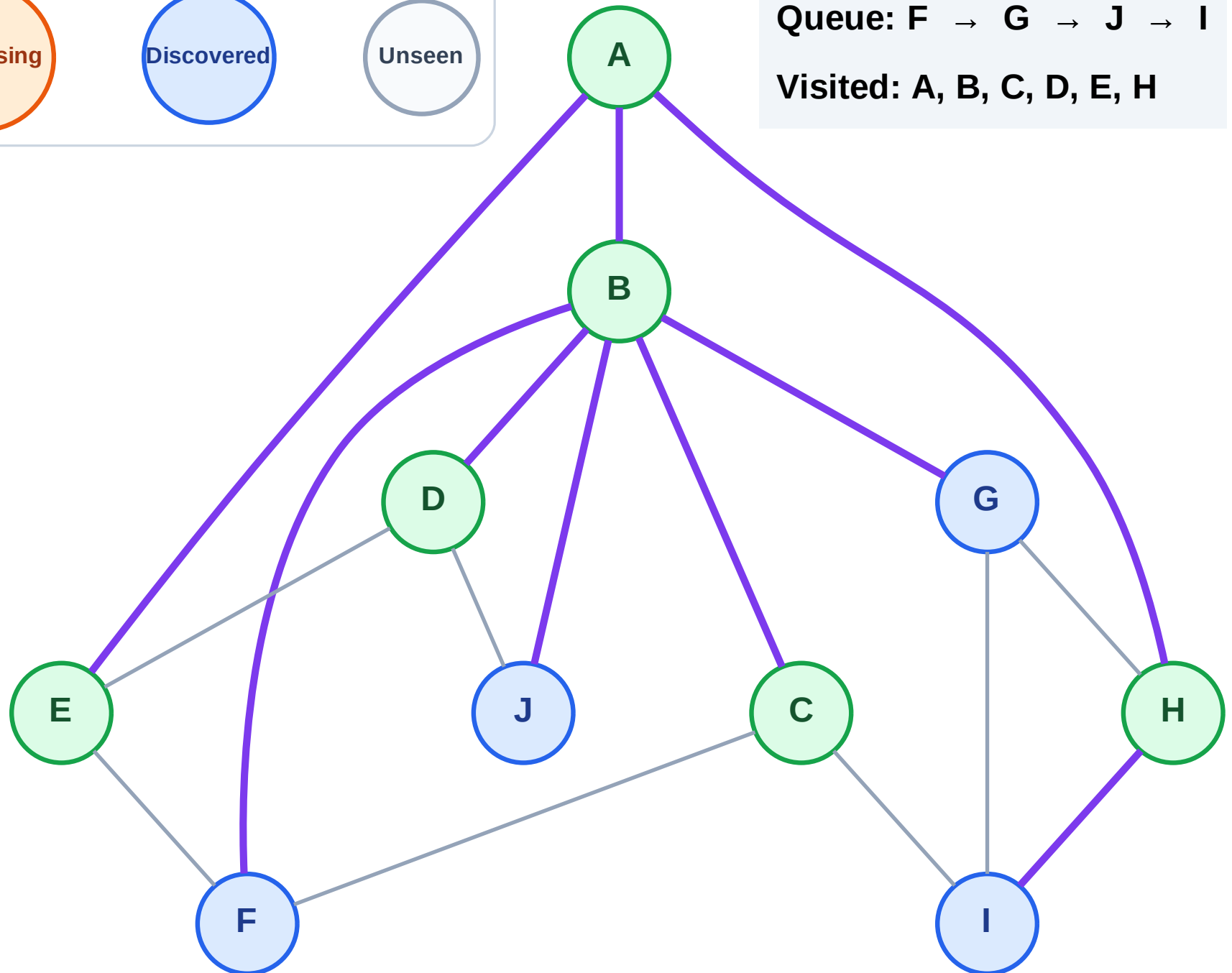
Processing

Discovered

Unseen

Queue: F → G → J → I

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node F

Legend

Complete

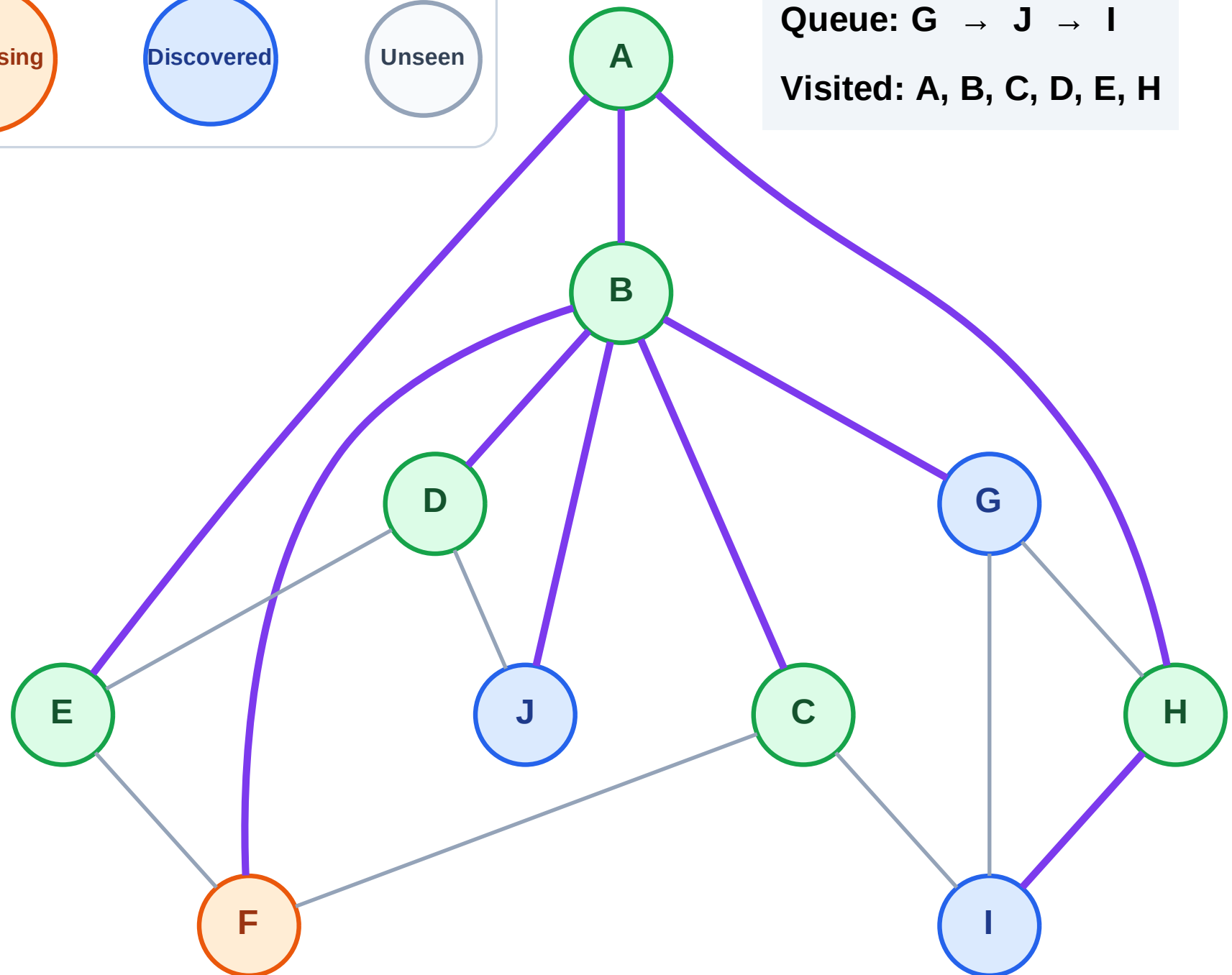
Processing

Discovered

Unseen

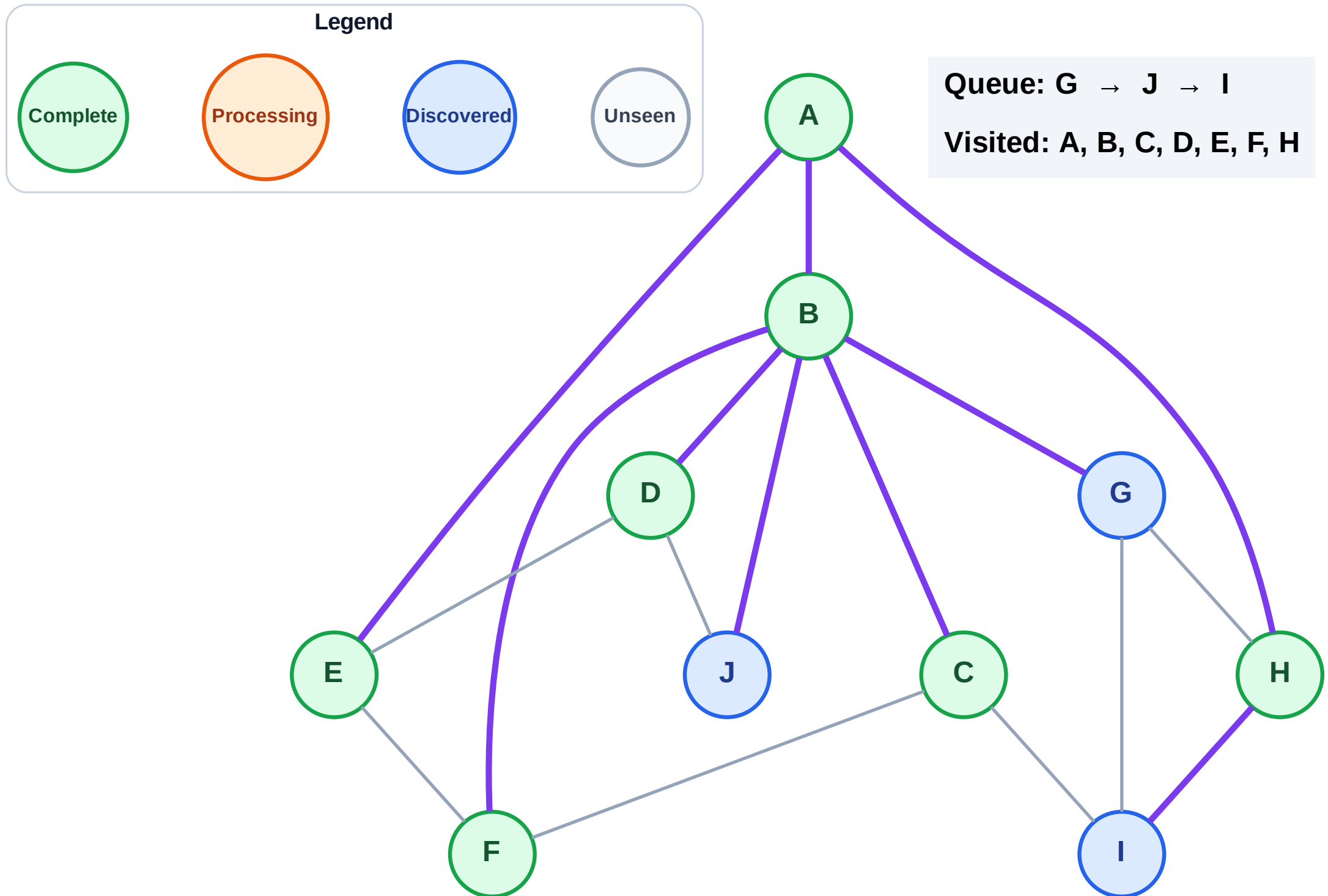
Queue: G → J → I

Visited: A, B, C, D, E, H



Step 15: No new neighbors from F

Step 15: No new neighbors from F



BFS Traversal

Step 16: Process node G

Legend

Complete

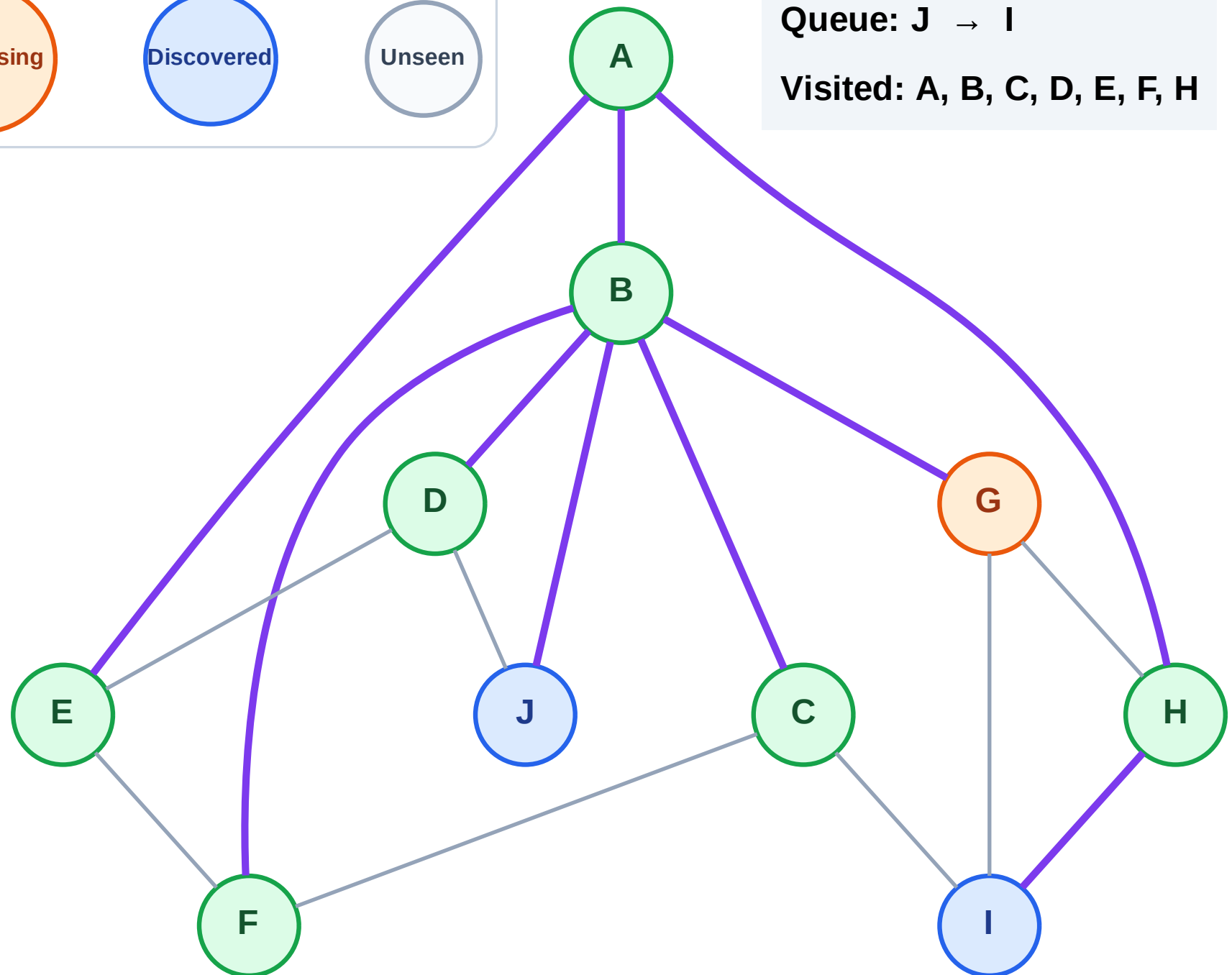
Processing

Discovered

Unseen

Queue: J → I

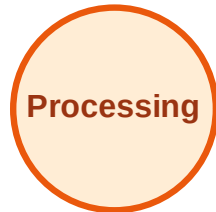
Visited: A, B, C, D, E, F, H



BFS Traversal

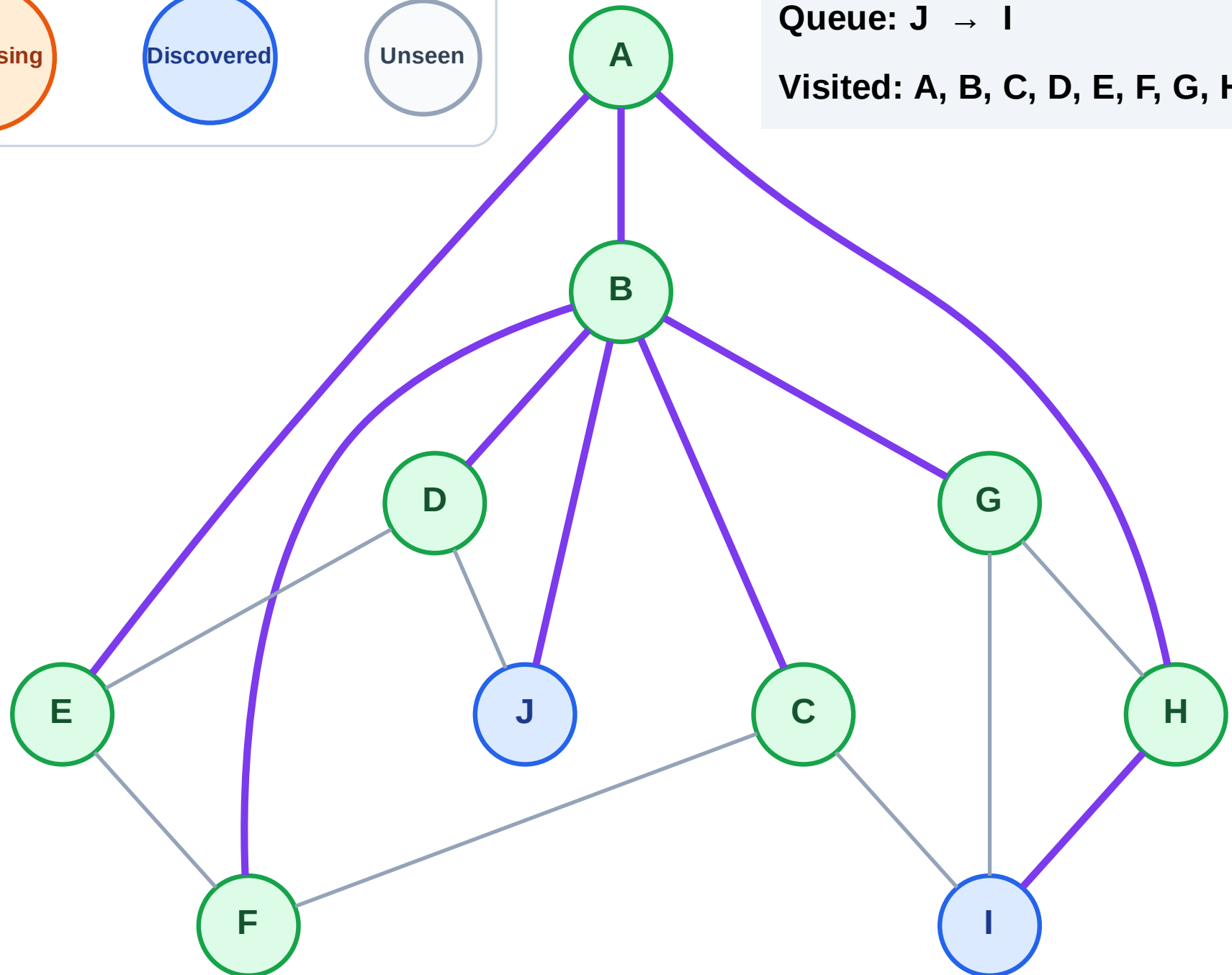
Step 17: No new neighbors from G

Legend



Queue: J → I

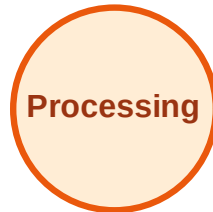
Visited: A, B, C, D, E, F, G, H



BFS Traversal

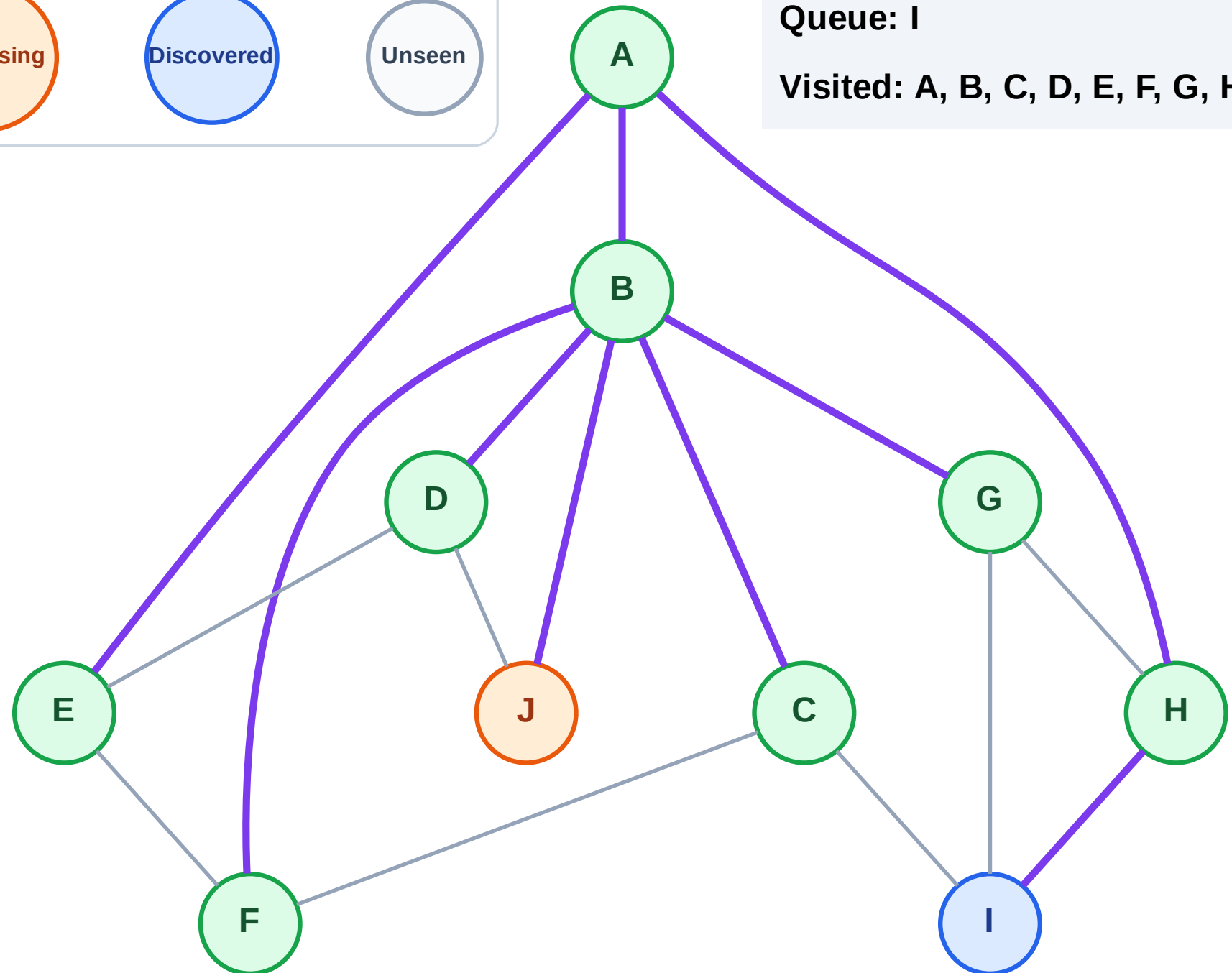
Step 18: Process node J

Legend



Queue: I

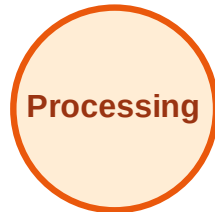
Visited: A, B, C, D, E, F, G, H



BFS Traversal

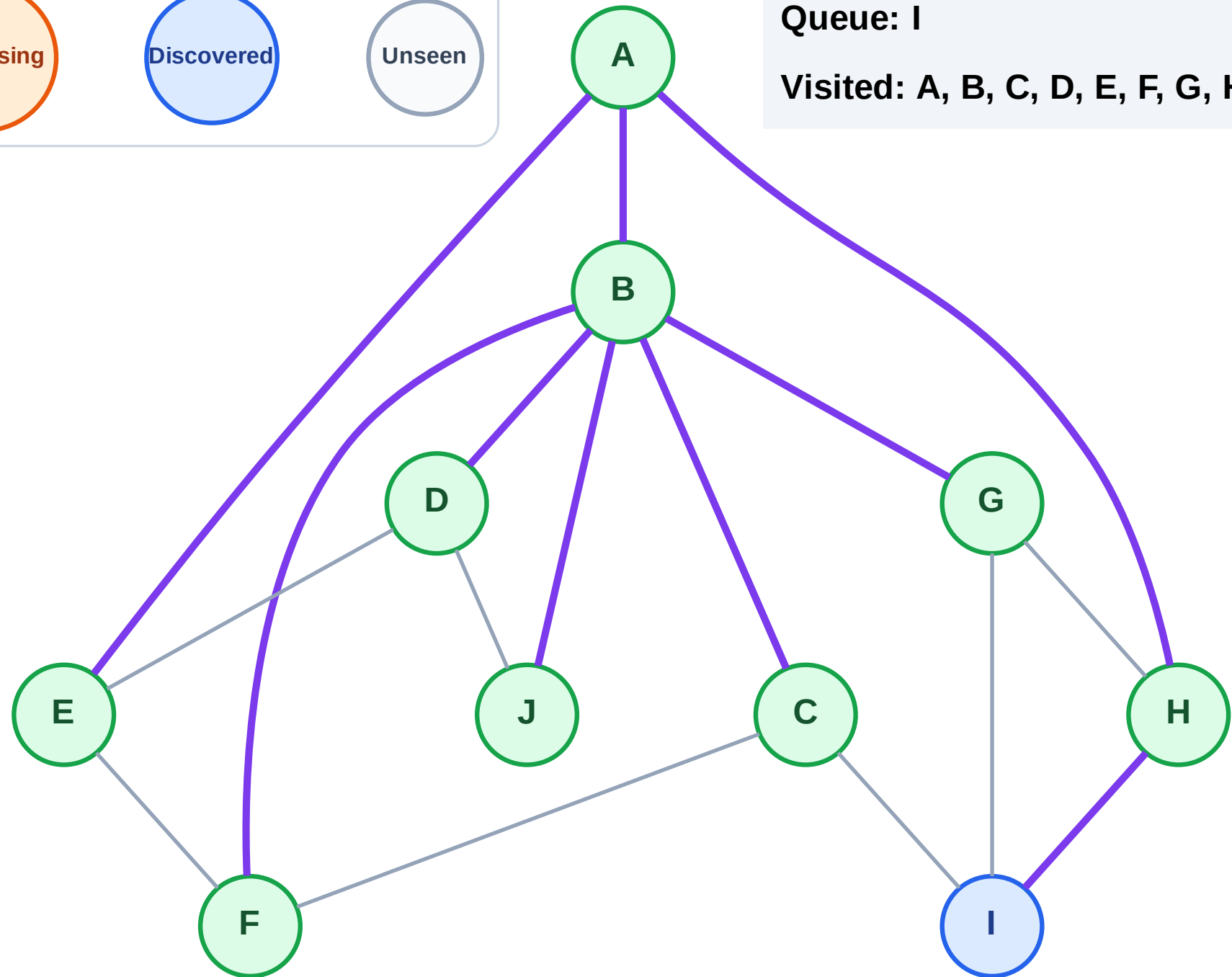
Step 19: No new neighbors from J

Legend



Queue: I

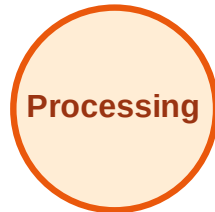
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

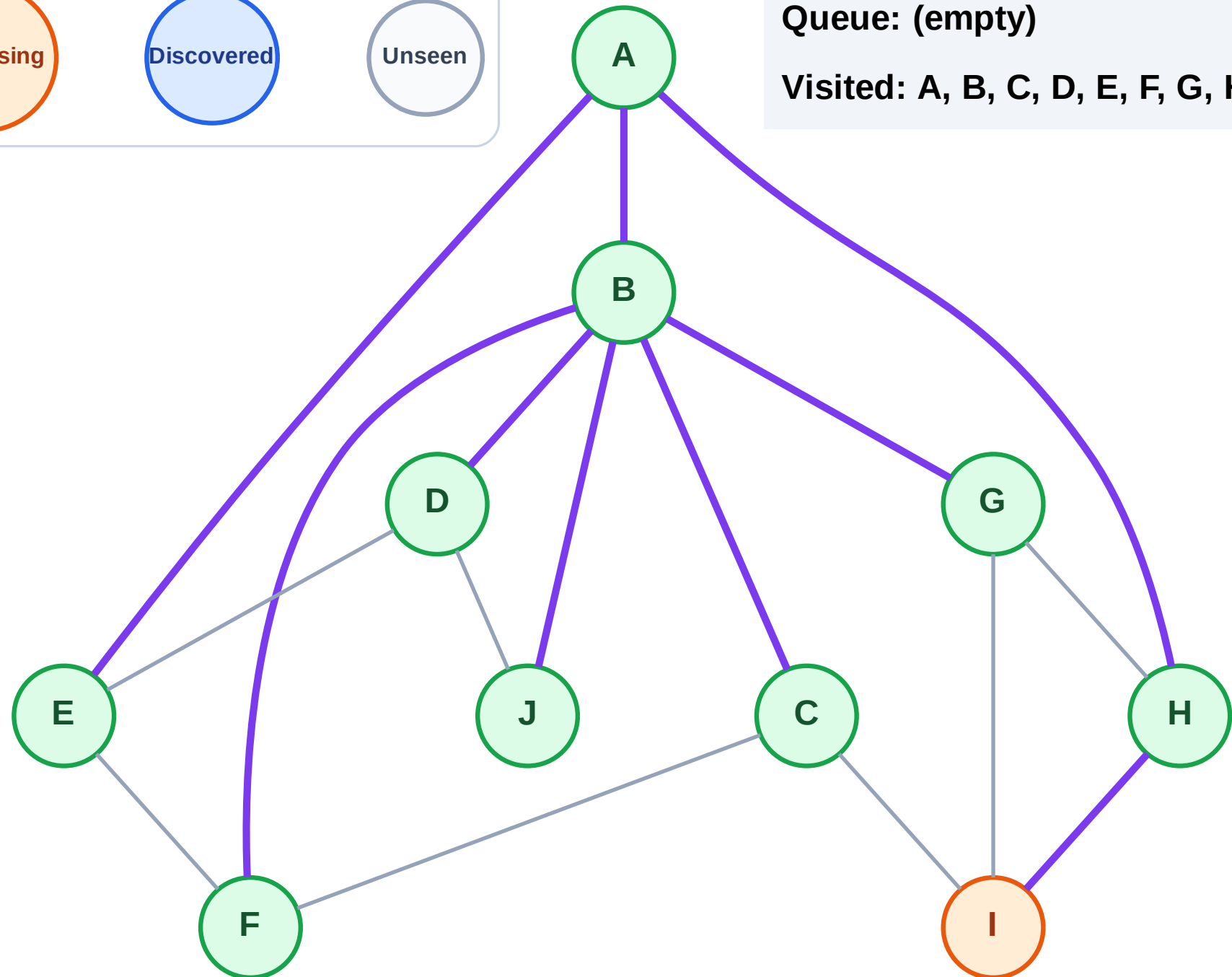
Step 20: Process node I

Legend



Queue: (empty)

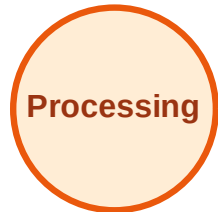
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

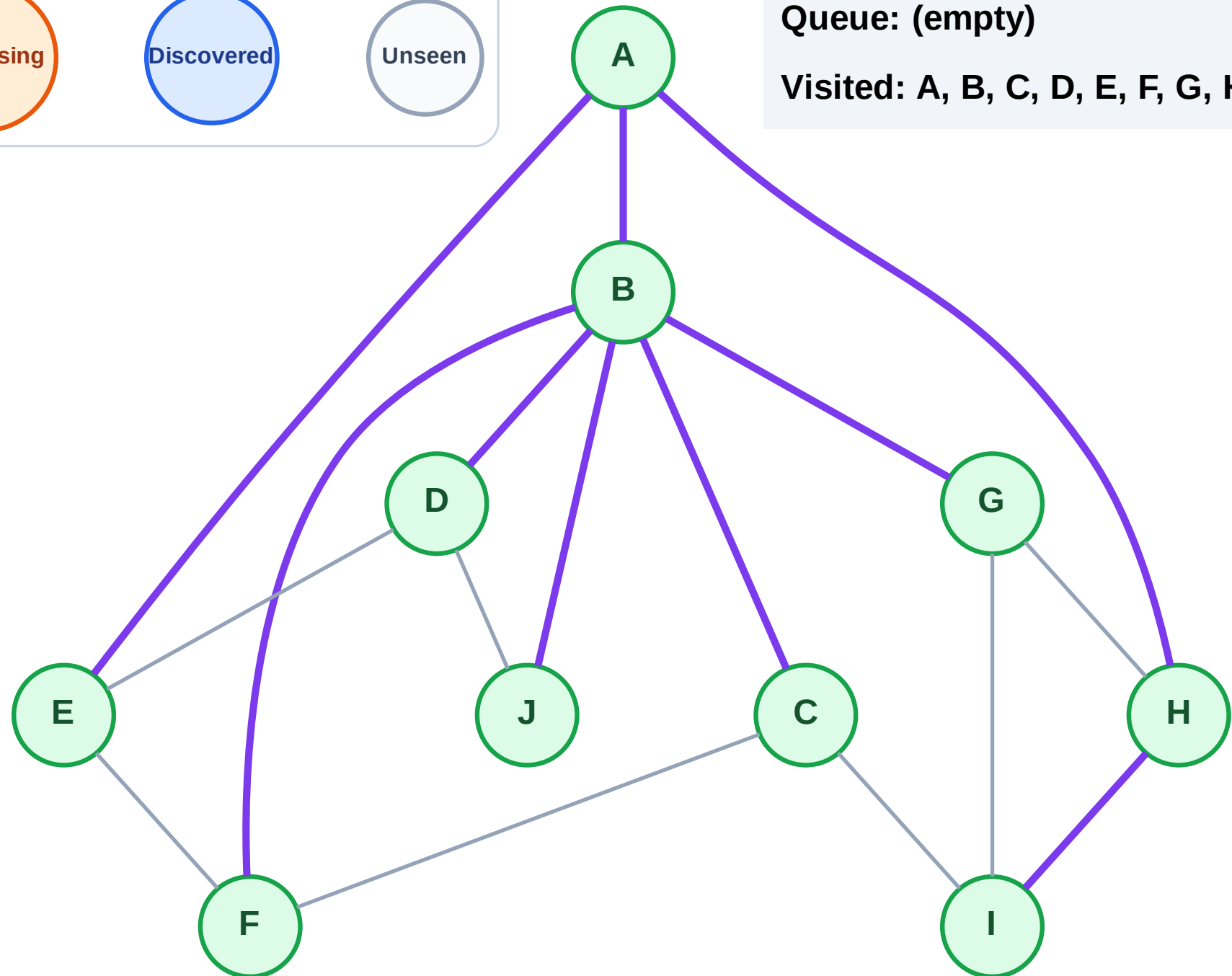
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

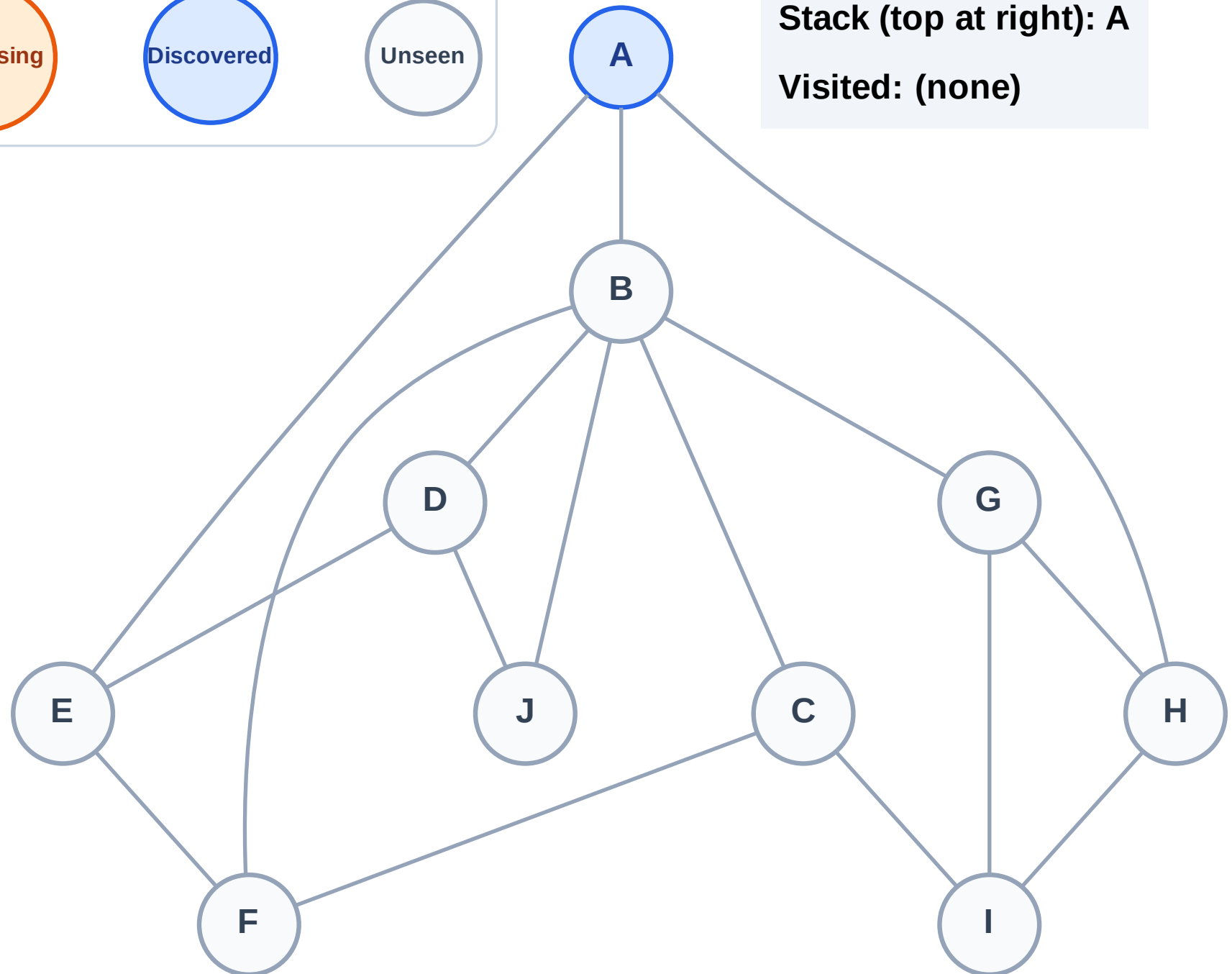
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

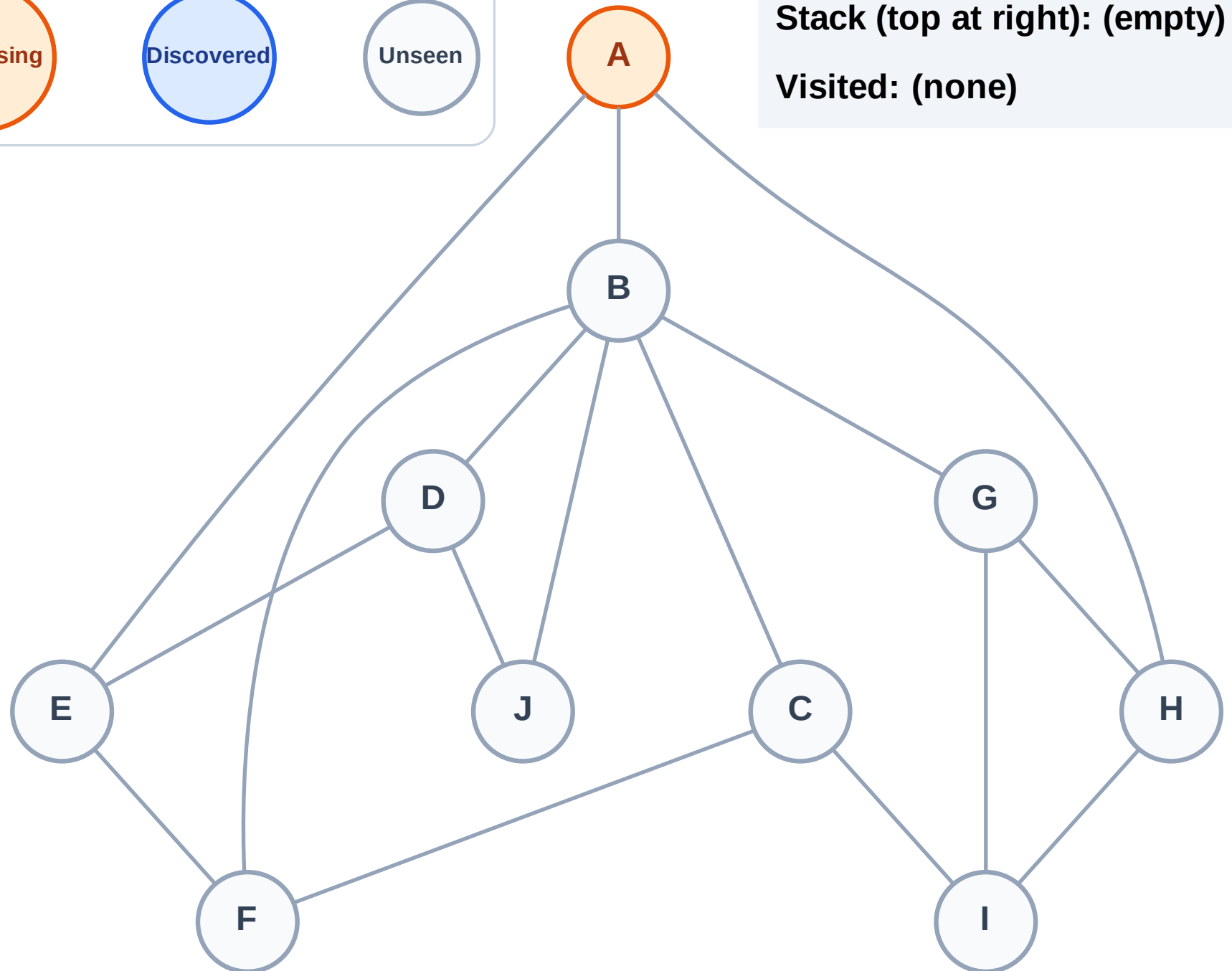
Processing

Discovered

Unseen

Stack (top at right): (empty)

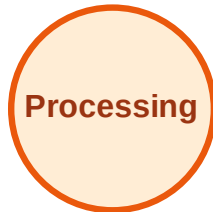
Visited: (none)



DFS Traversal

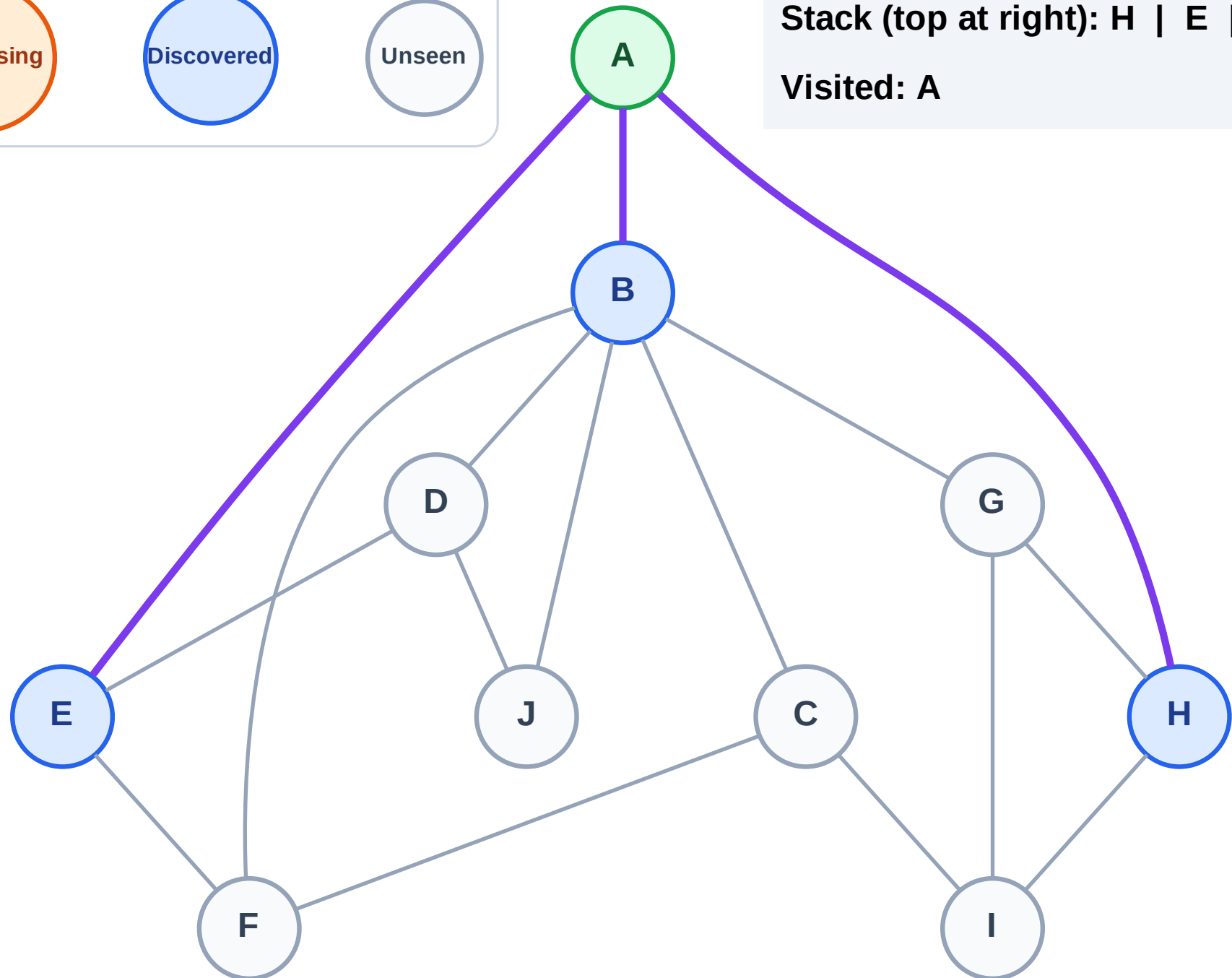
Step 3: Discover H, E, B from A

Legend



Stack (top at right): H | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

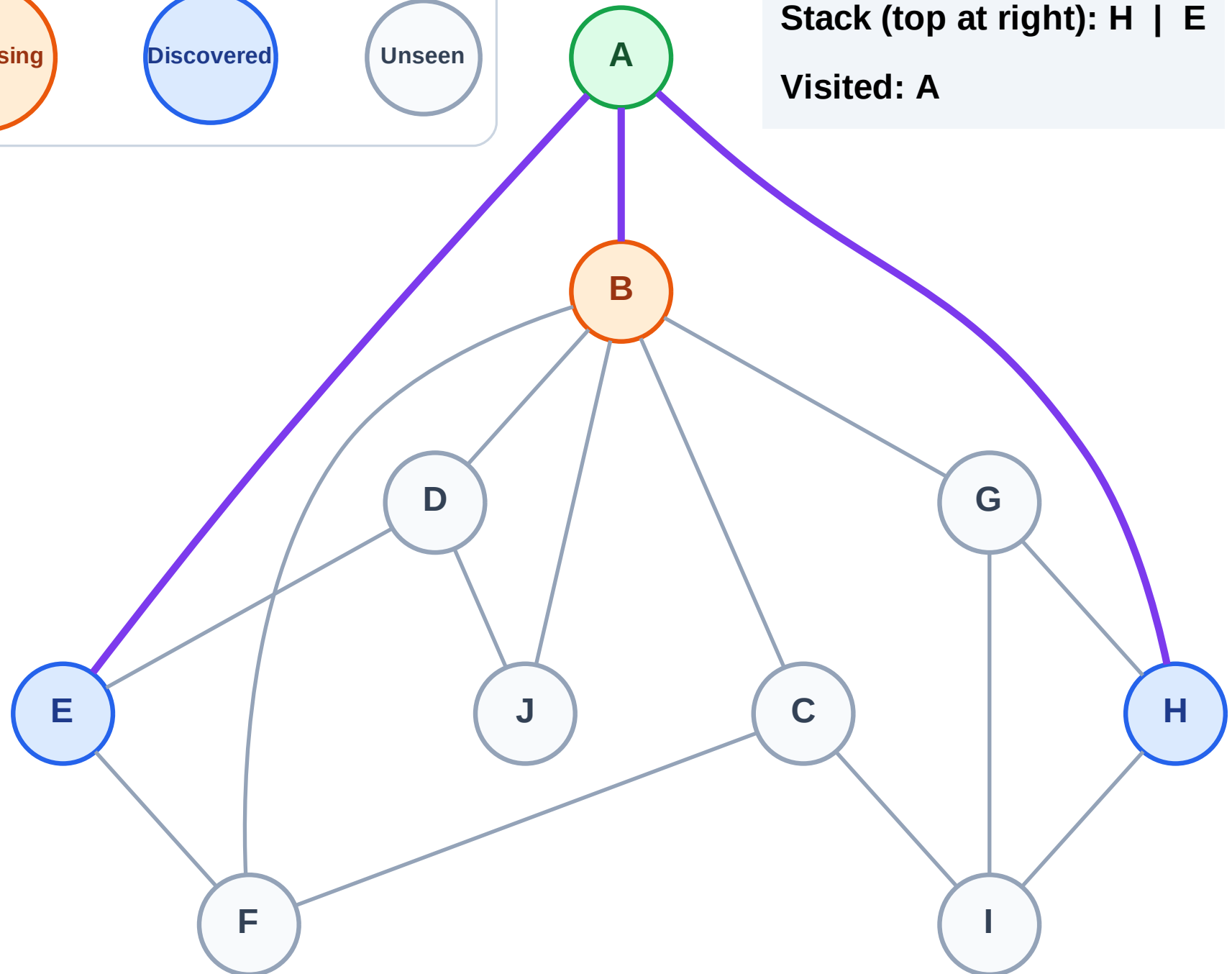
Processing

Discovered

Unseen

Stack (top at right): H | E

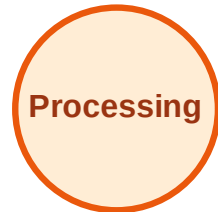
Visited: A



DFS Traversal

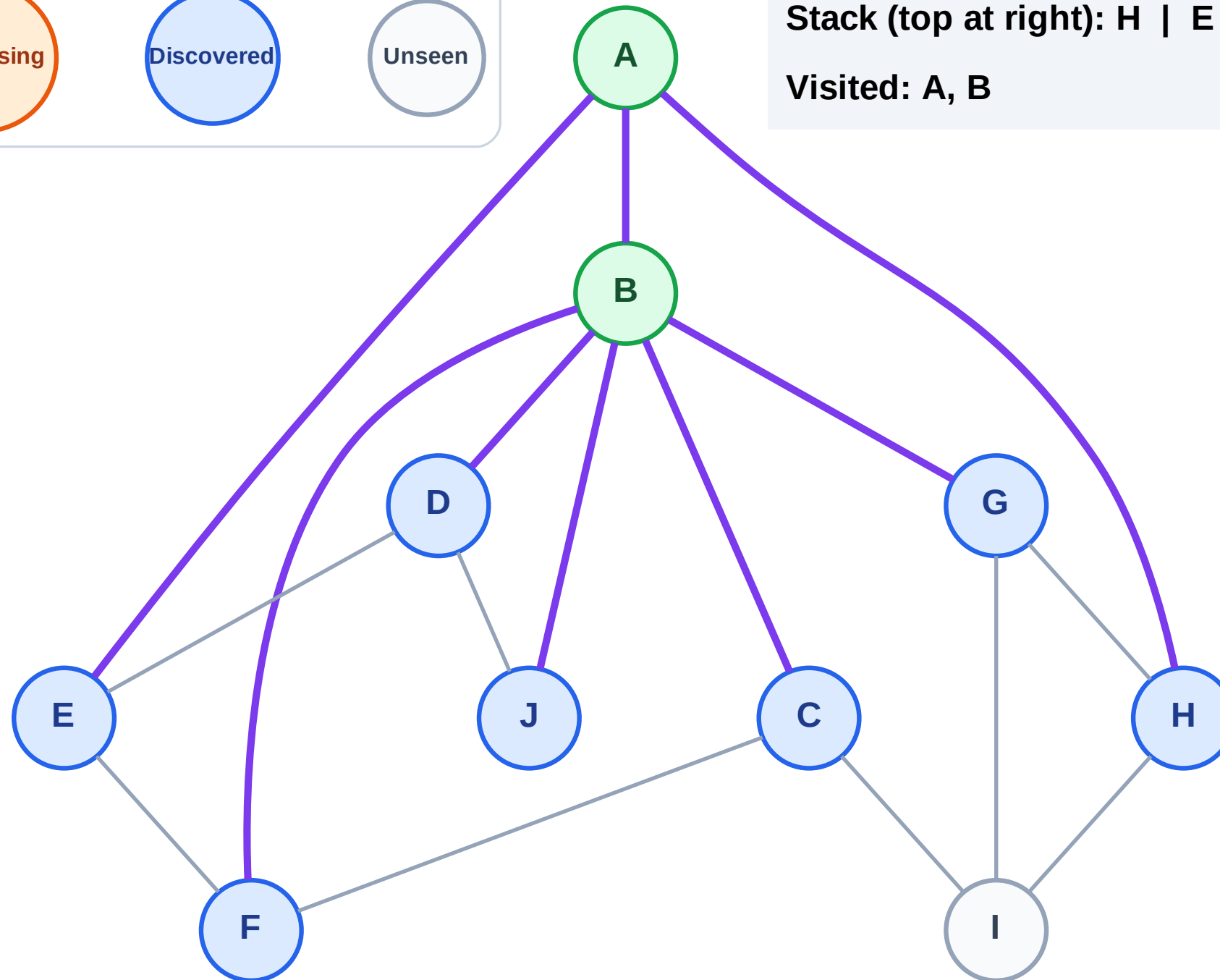
Step 5: Discover J, G, F, D, C from B

Legend



Stack (top at right): H | E | J | G | F | D | C

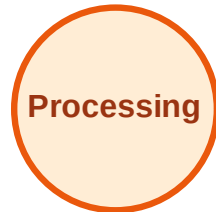
Visited: A, B



DFS Traversal

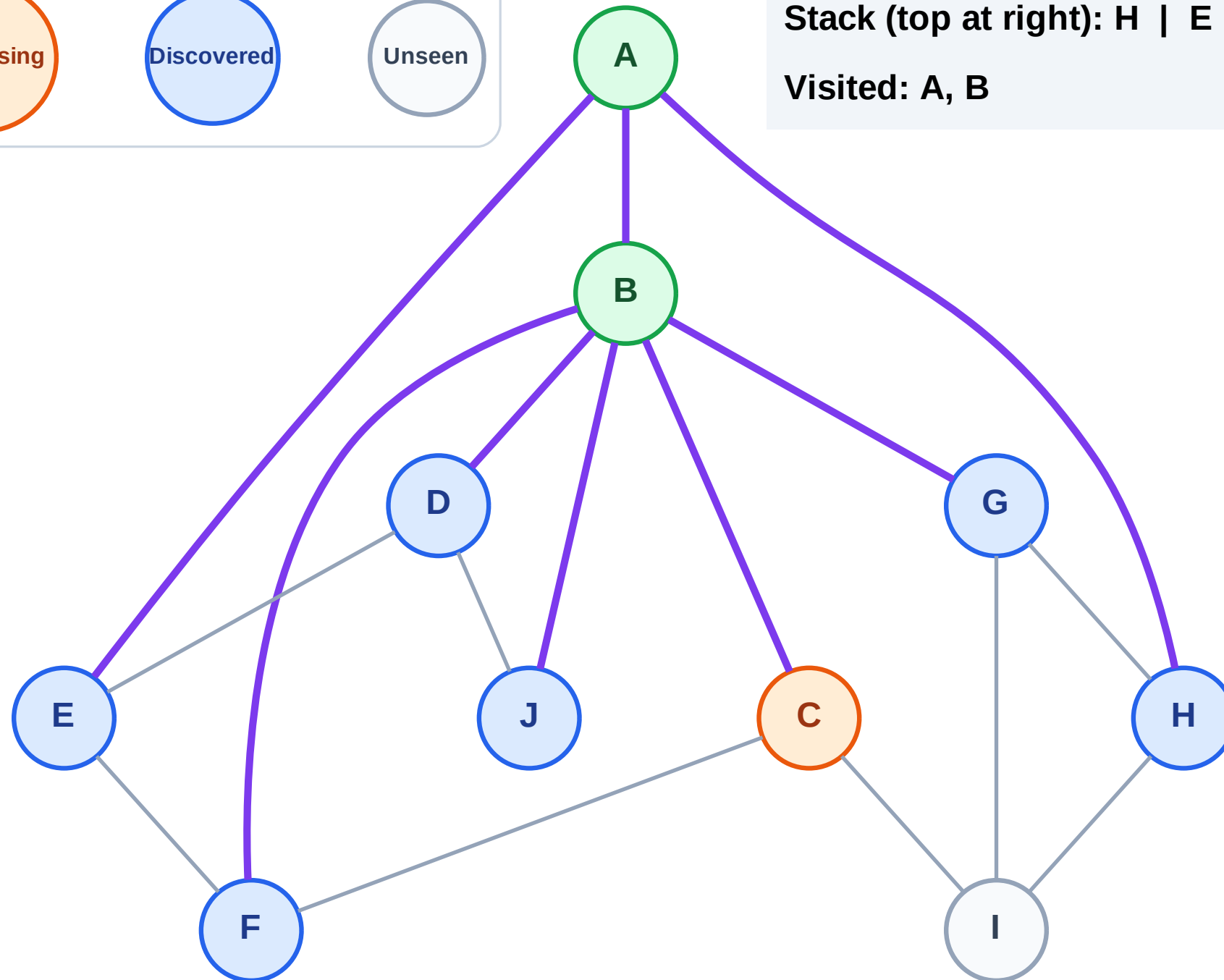
Step 6: Process node C

Legend



Stack (top at right): H | E | J | G | F | D

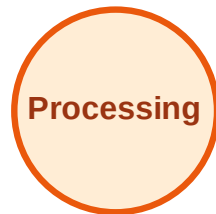
Visited: A, B



DFS Traversal

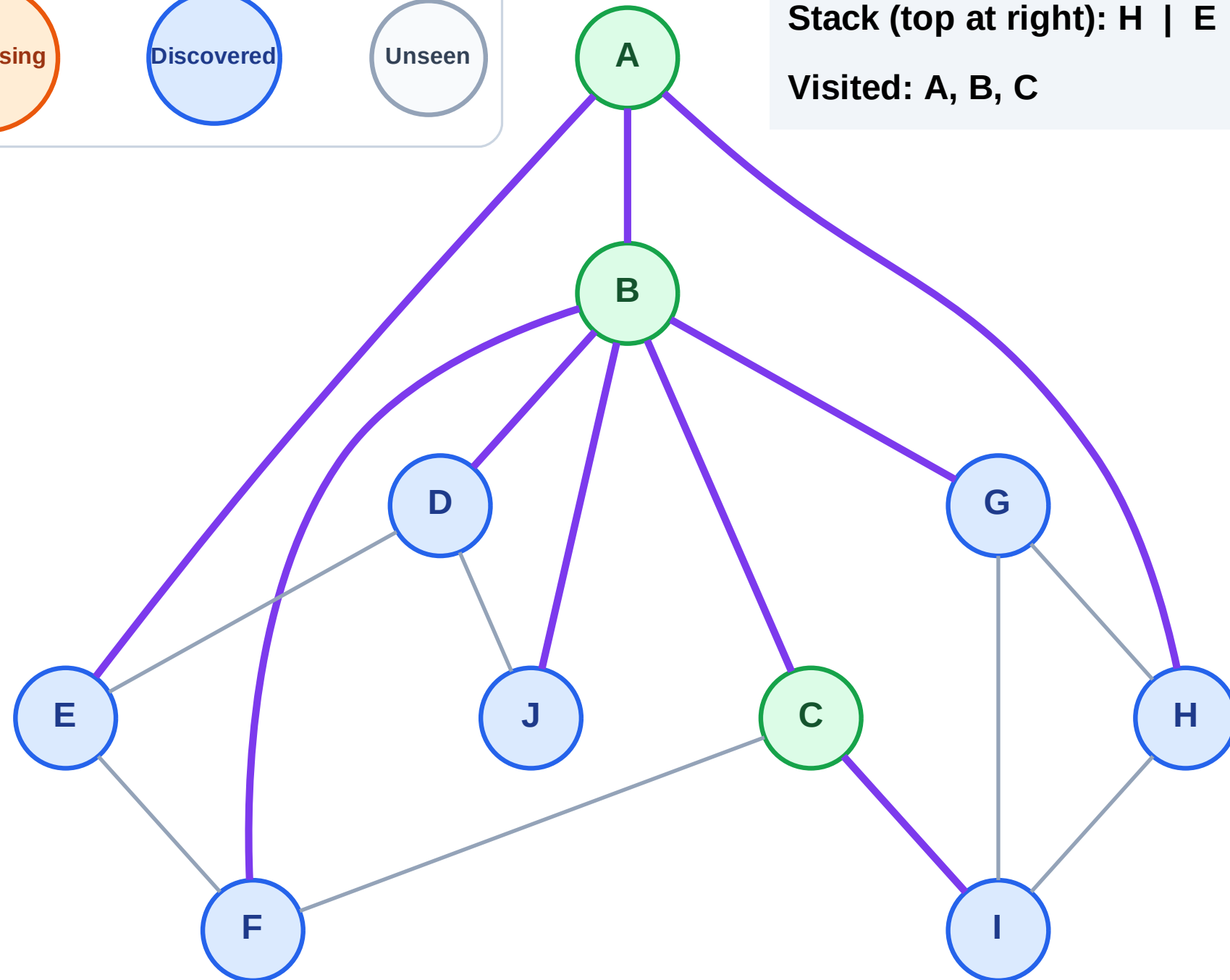
Step 7: Discover I from C

Legend



Stack (top at right): H | E | J | G | F | D | I

Visited: A, B, C



DFS Traversal

Step 8: Process node I

Legend

Complete

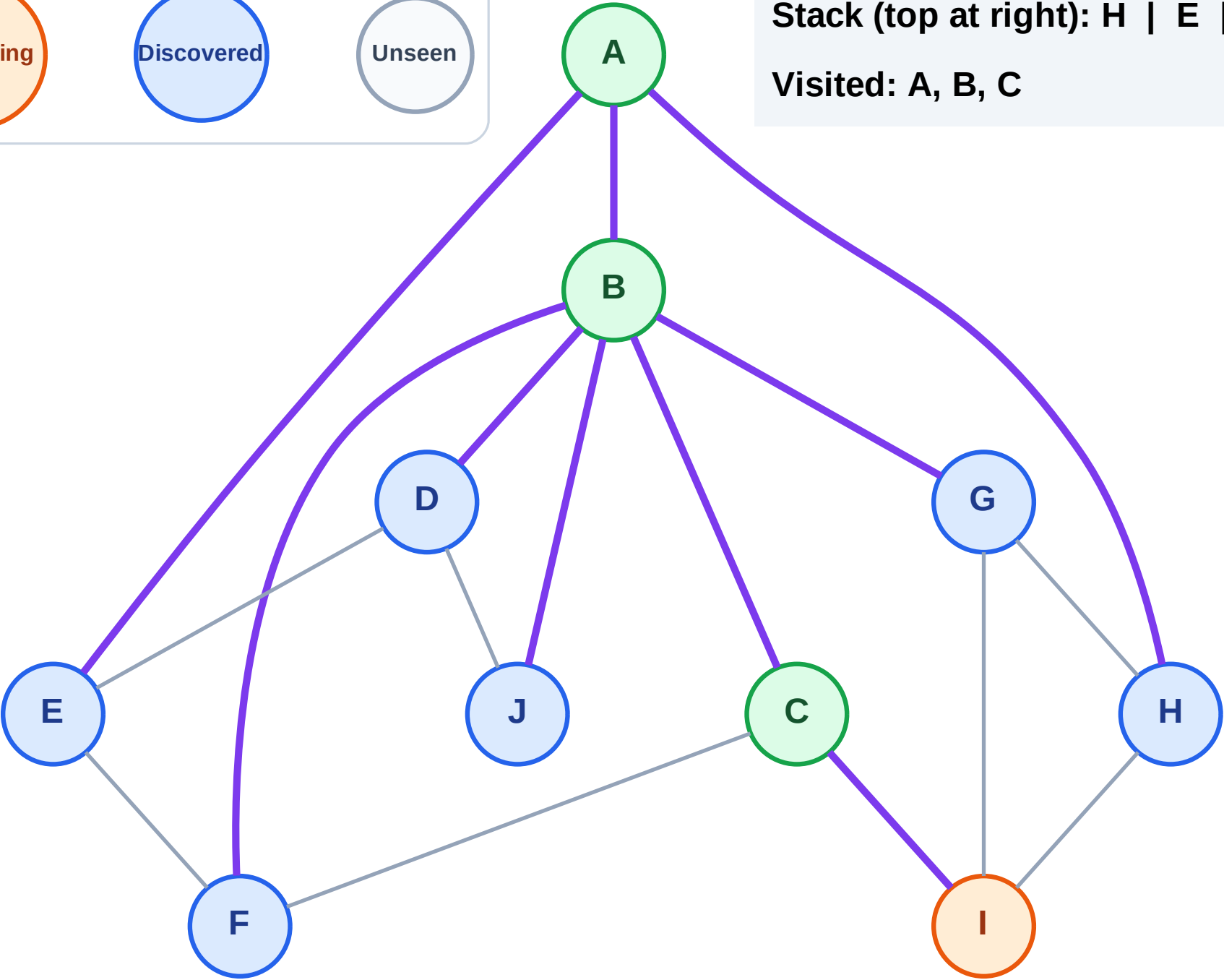
Processing

Discovered

Unseen

Stack (top at right): H | E | J | G | F | D

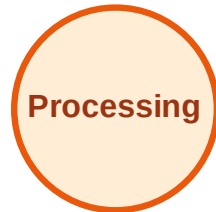
Visited: A, B, C



DFS Traversal

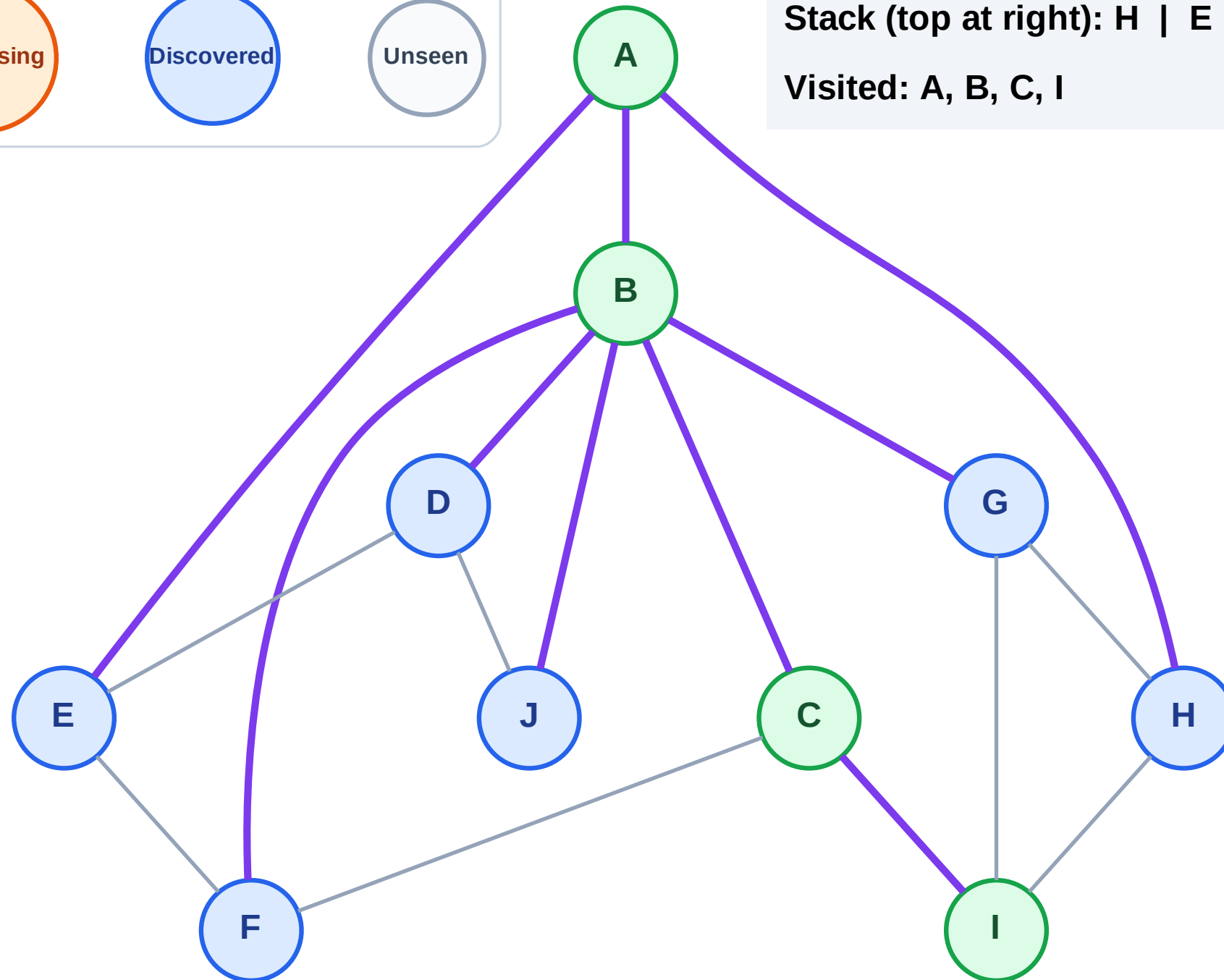
Step 9: No new neighbors from I

Legend



Stack (top at right): H | E | J | G | F | D

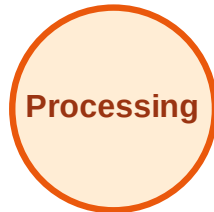
Visited: A, B, C, I



DFS Traversal

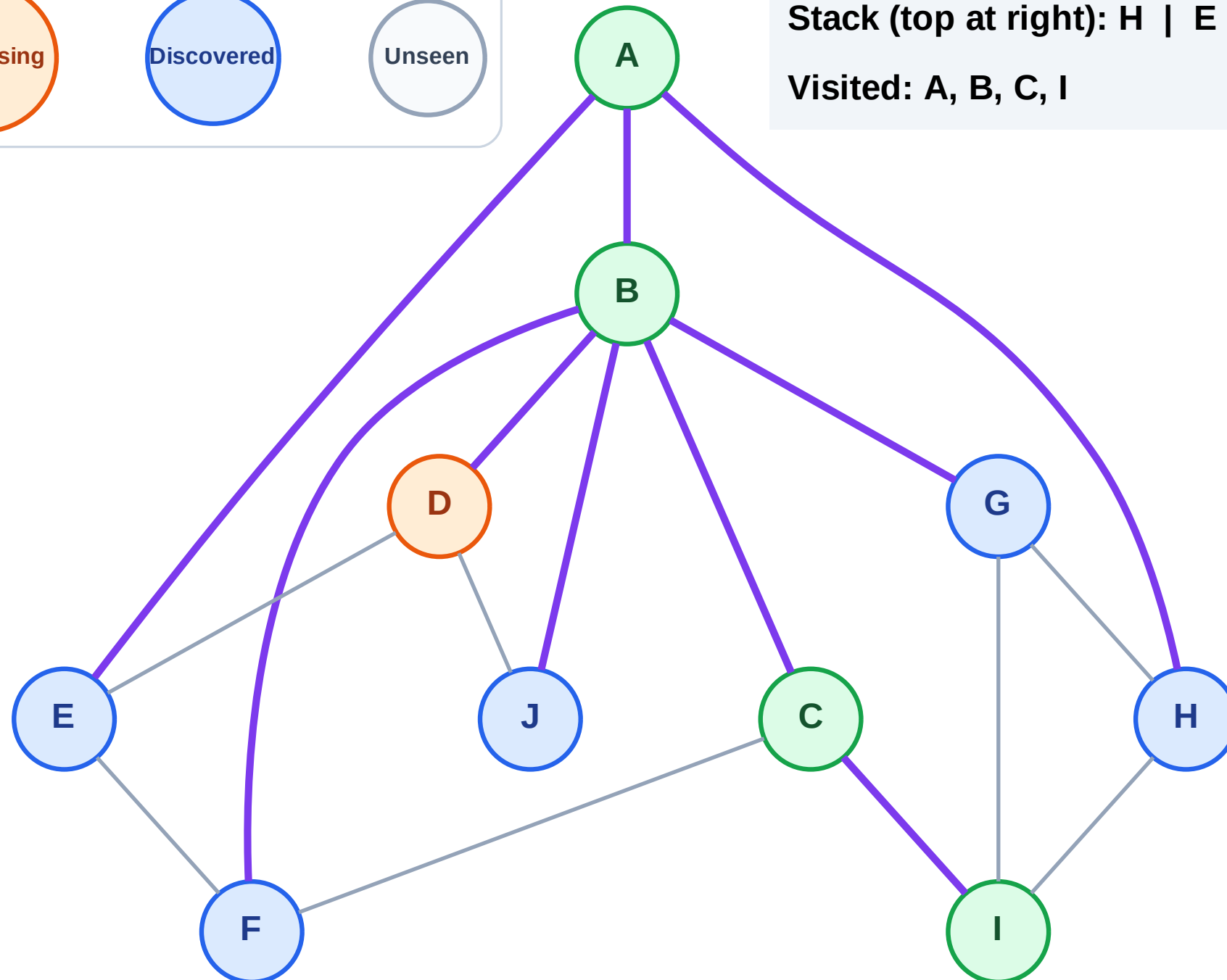
Step 10: Process node D

Legend



Stack (top at right): H | E | J | G | F

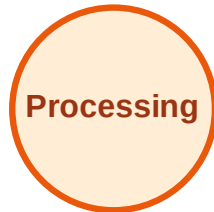
Visited: A, B, C, I



DFS Traversal

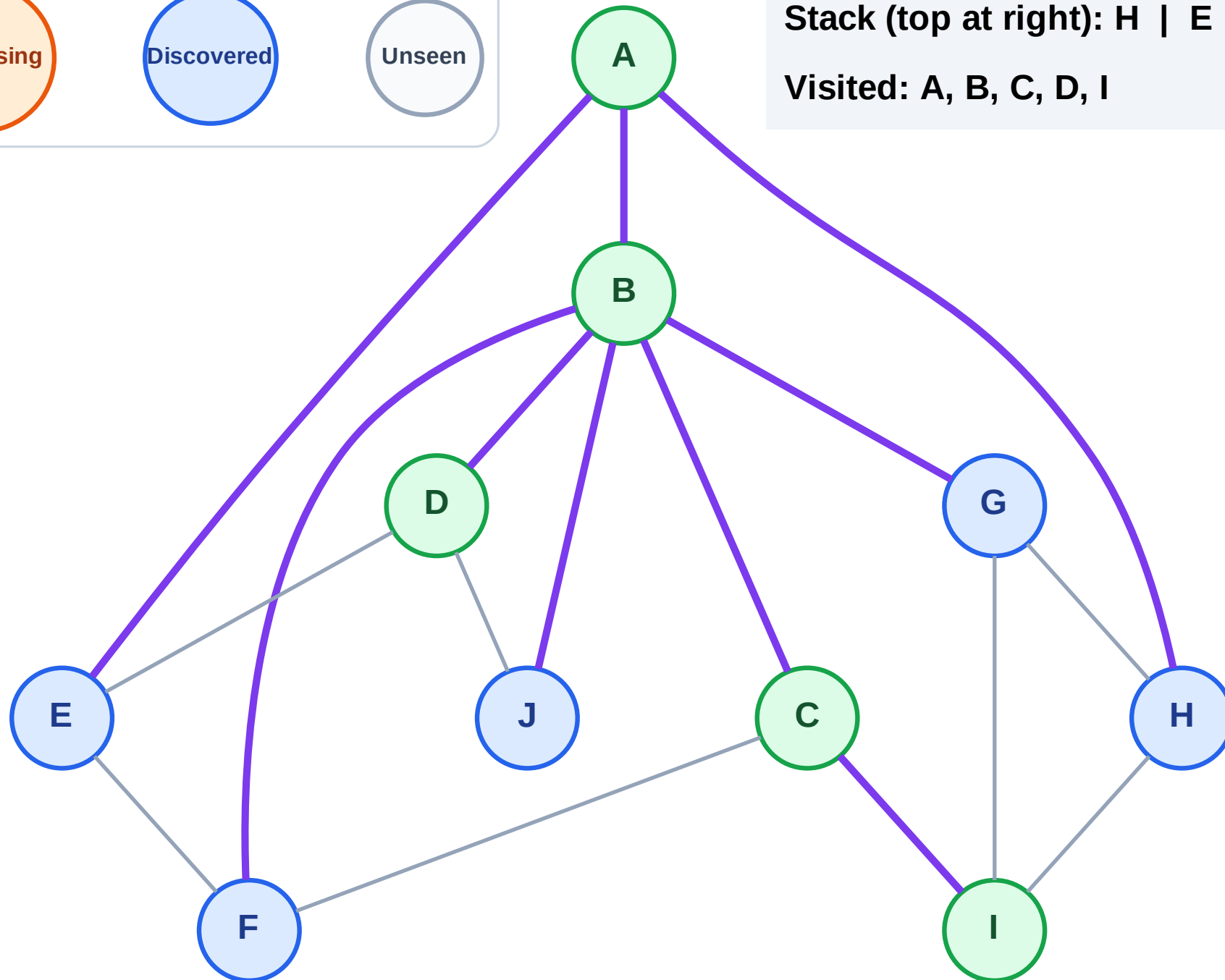
Step 11: No new neighbors from D

Legend



Stack (top at right): H | E | J | G | F

Visited: A, B, C, D, I



DFS Traversal

Step 12: Process node F

Legend

Complete

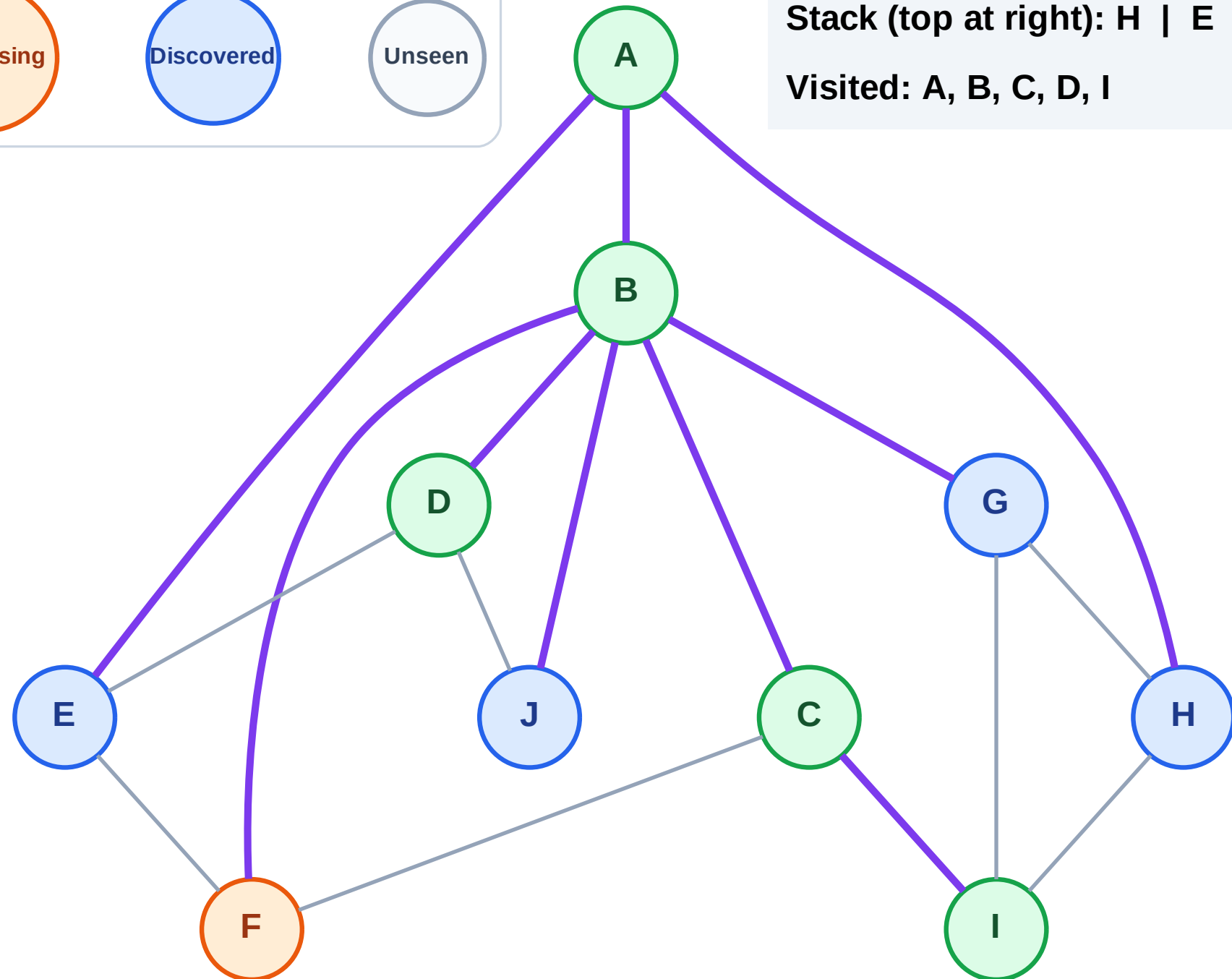
Processing

Discovered

Unseen

Stack (top at right): H | E | J | G

Visited: A, B, C, D, I



DFS Traversal

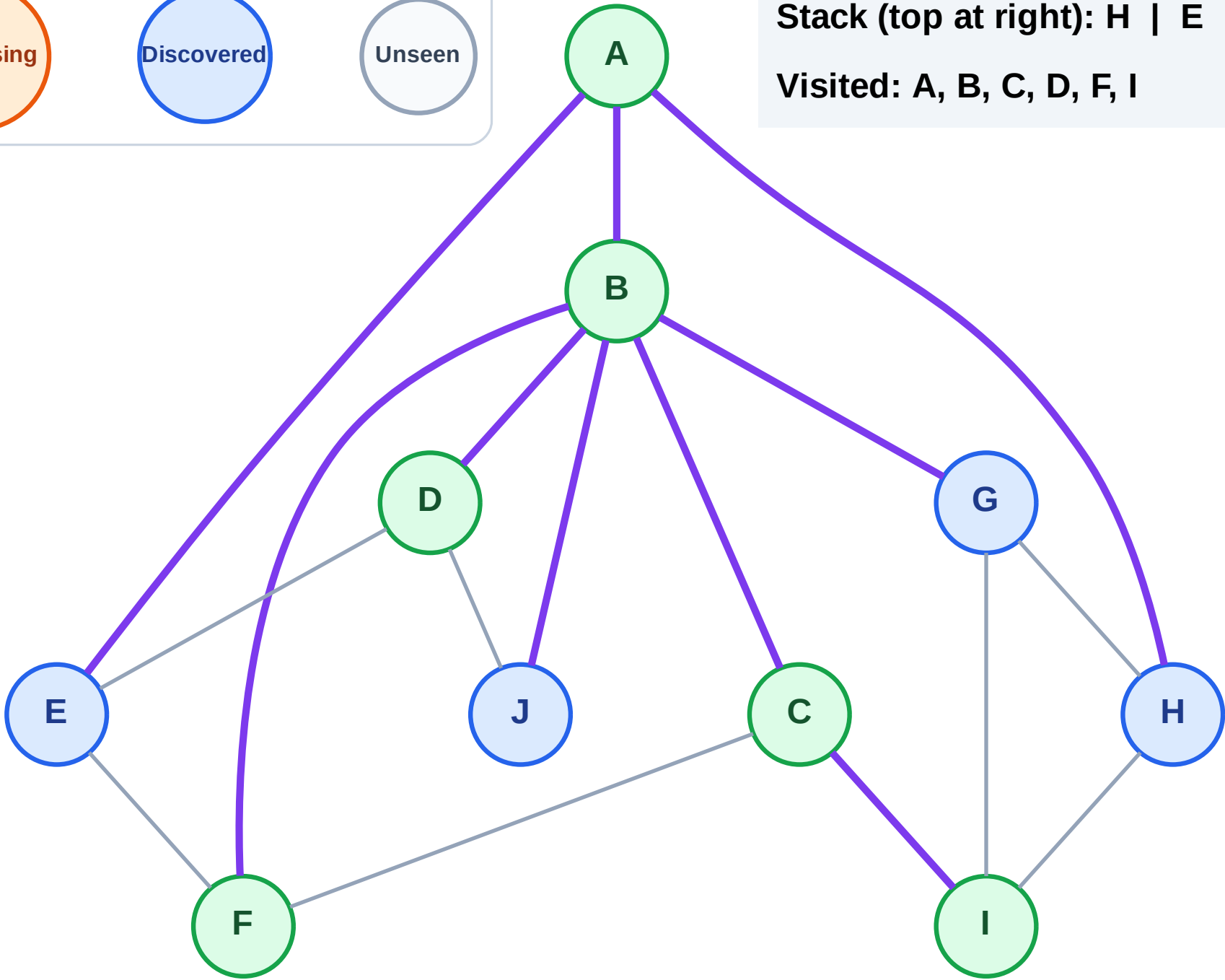
Step 13: No new neighbors from F

Legend



Stack (top at right): H | E | J | G

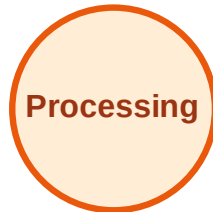
Visited: A, B, C, D, F, I



DFS Traversal

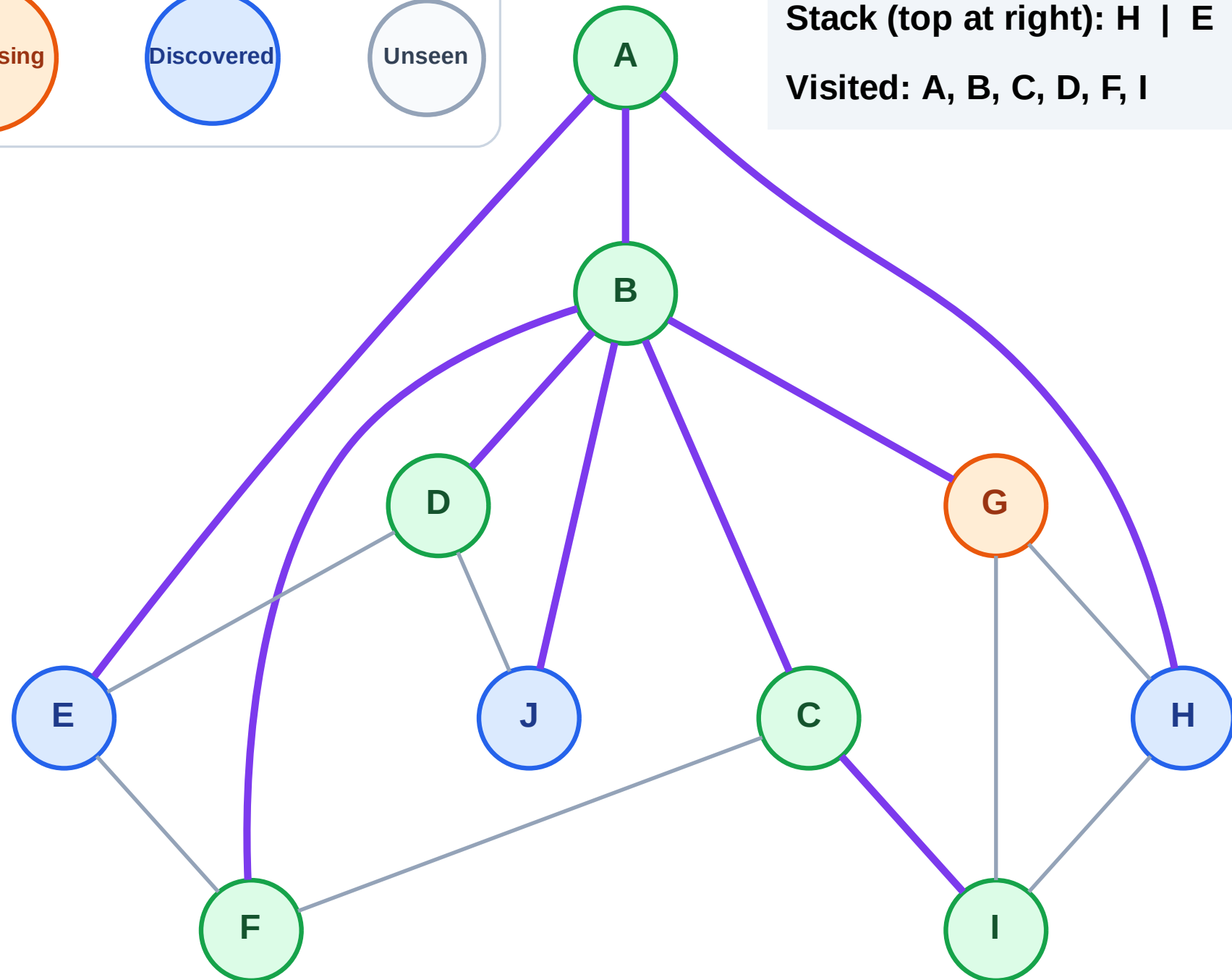
Step 14: Process node G

Legend



Stack (top at right): H | E | J

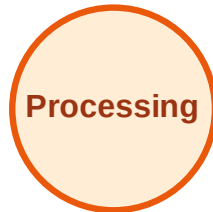
Visited: A, B, C, D, F, I



DFS Traversal

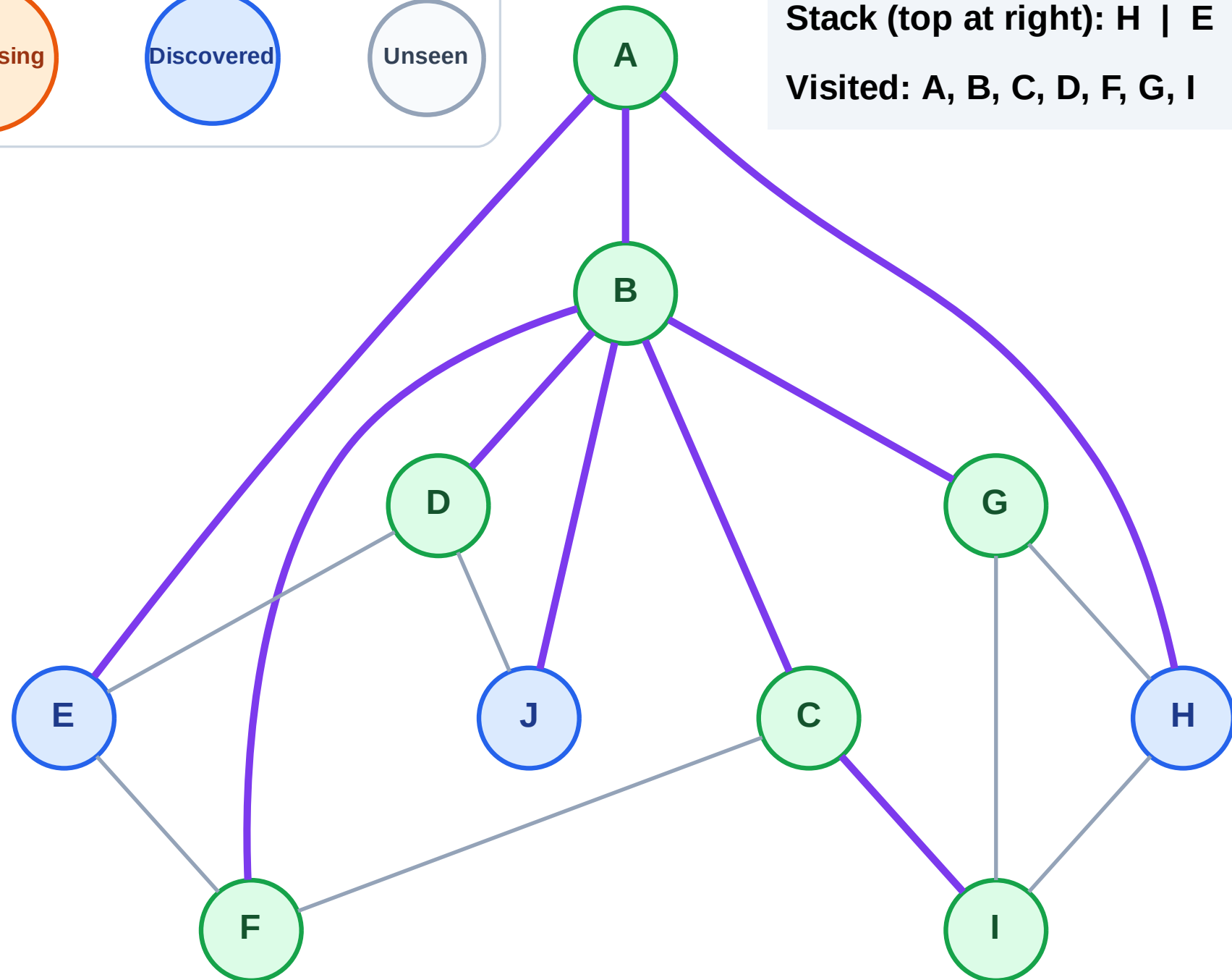
Step 15: No new neighbors from G

Legend



Stack (top at right): H | E | J

Visited: A, B, C, D, F, G, I



DFS Traversal

Step 16: Process node J

Legend

Complete

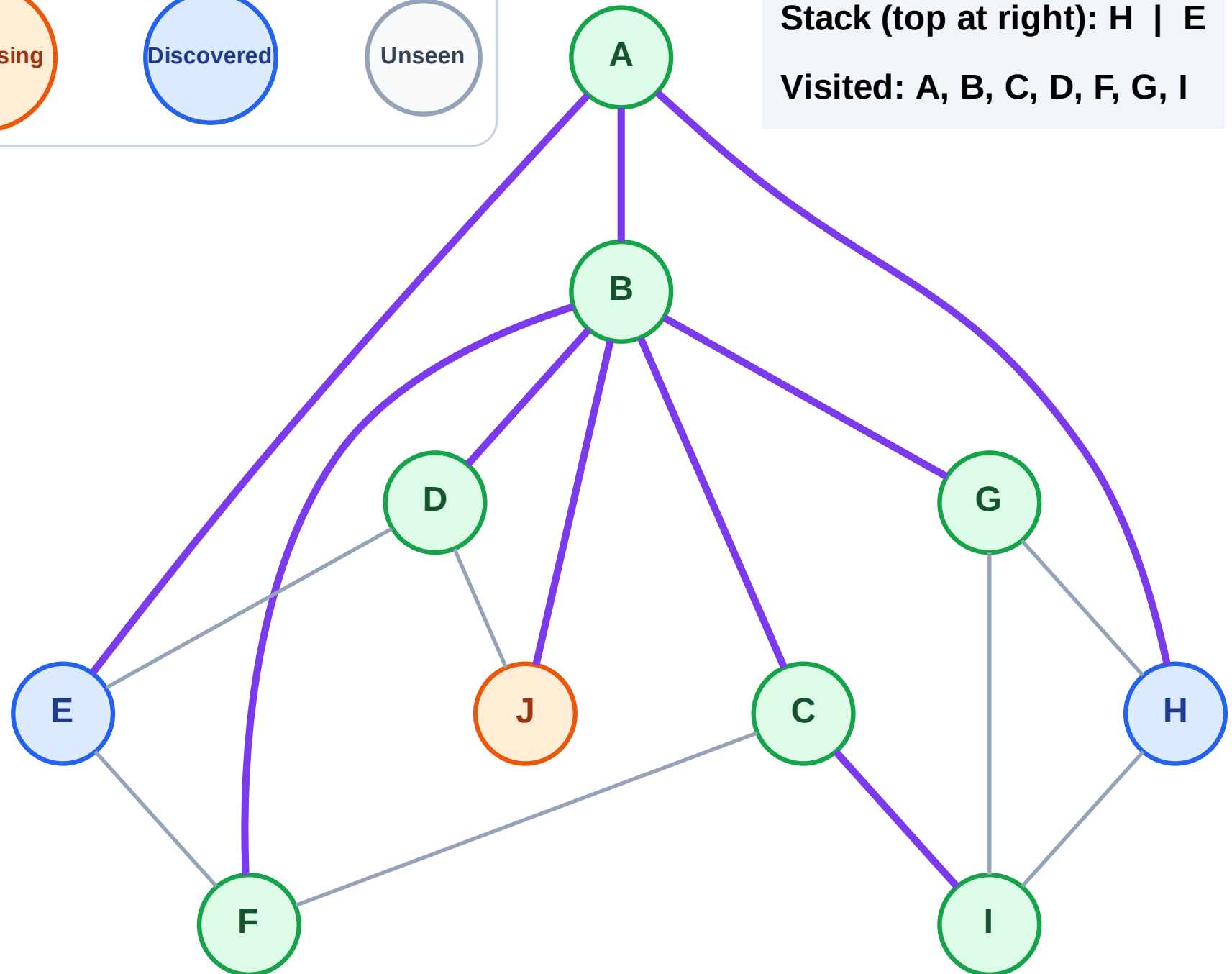
Processing

Discovered

Unseen

Stack (top at right): H | E

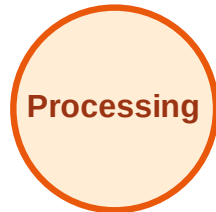
Visited: A, B, C, D, F, G, I



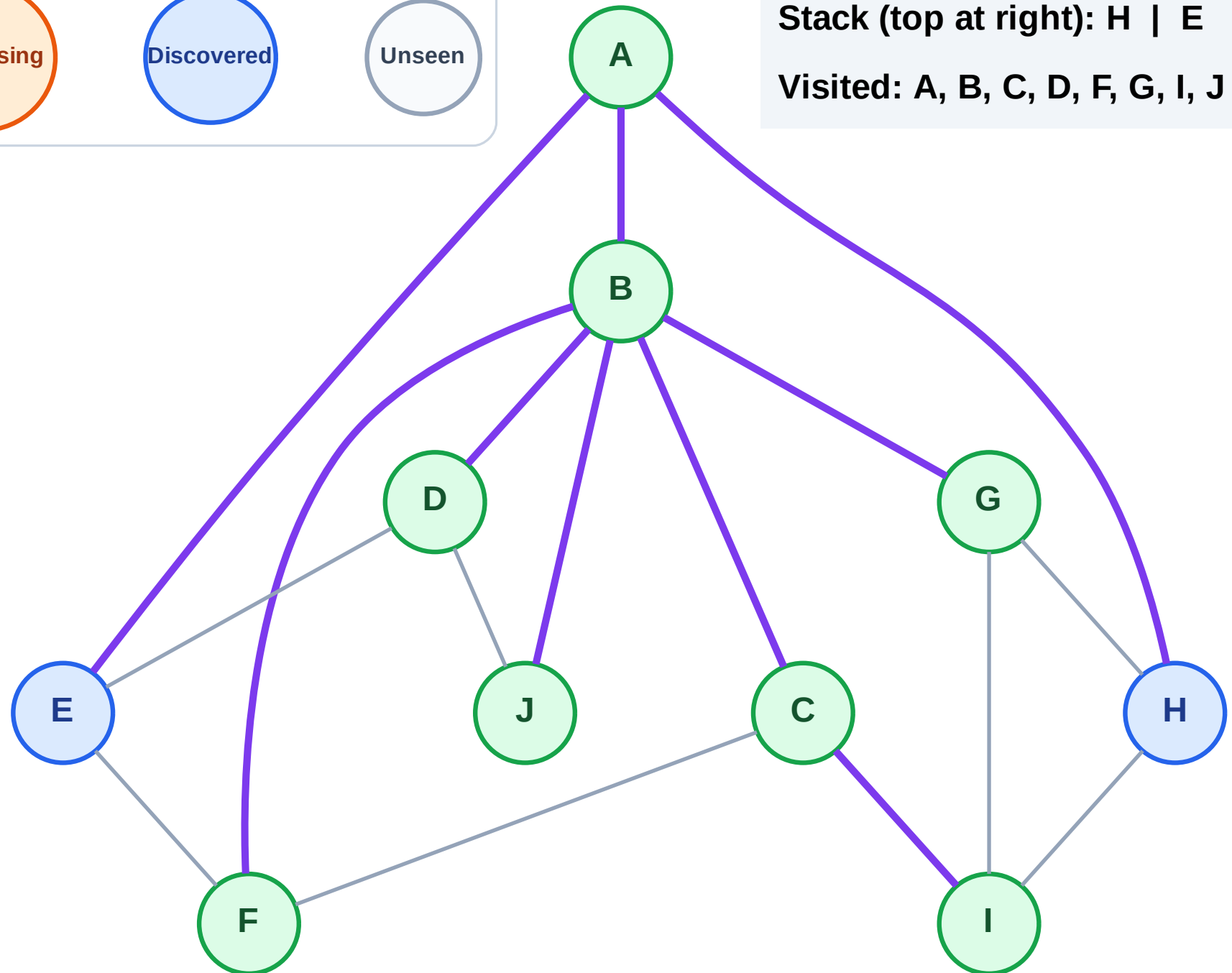
DFS Traversal

Step 17: No new neighbors from J

Legend

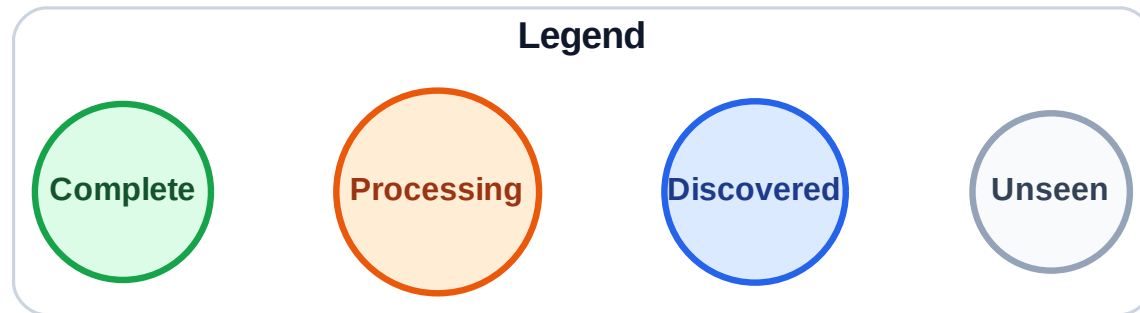


Stack (top at right): H | E
Visited: A, B, C, D, F, G, I, J



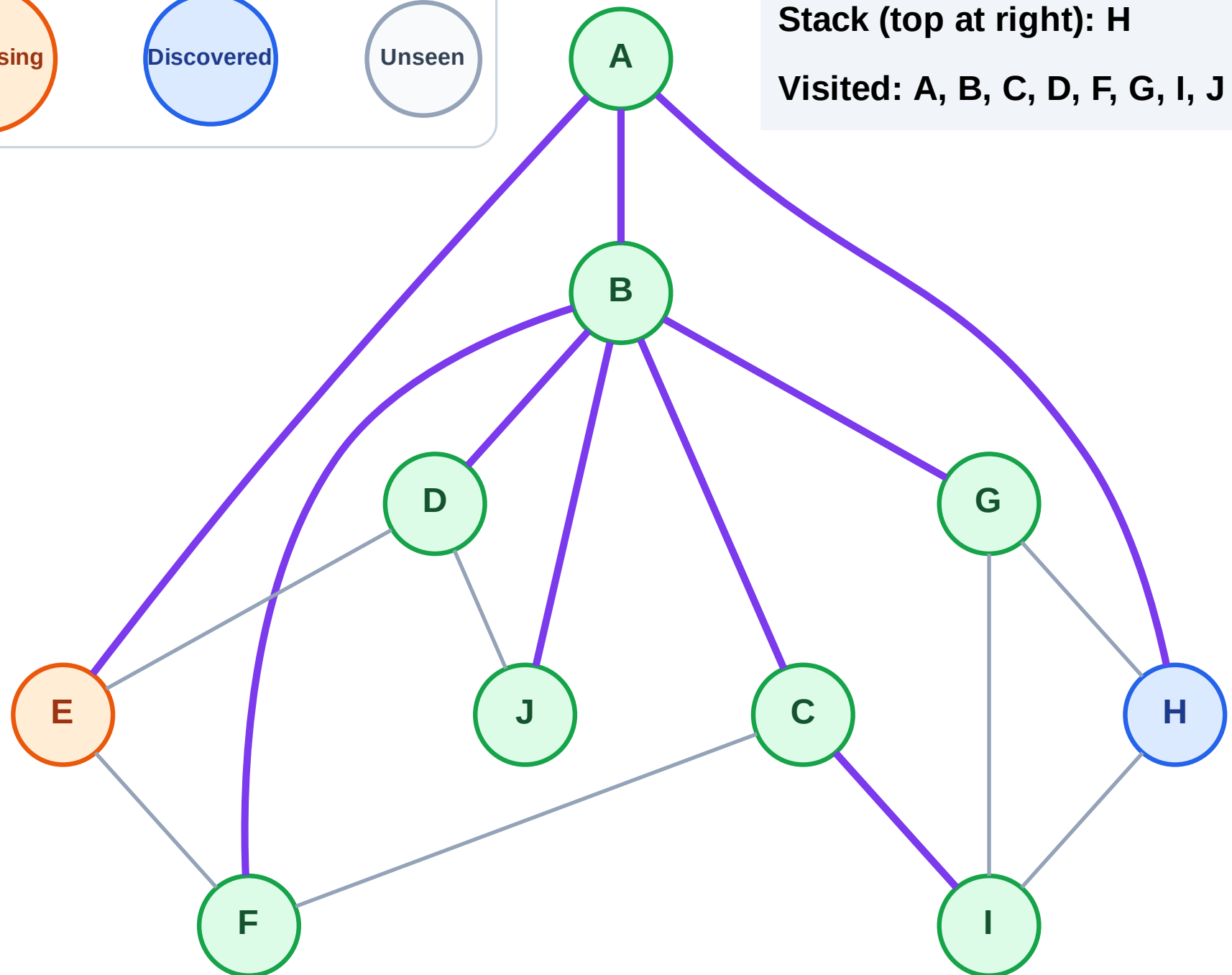
Step 18: Process node E

Step 18: Process node E



Stack (top at right): H

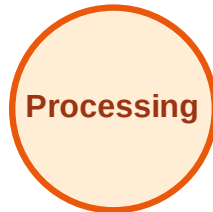
Visited: A, B, C, D, F, G, I, J



DFS Traversal

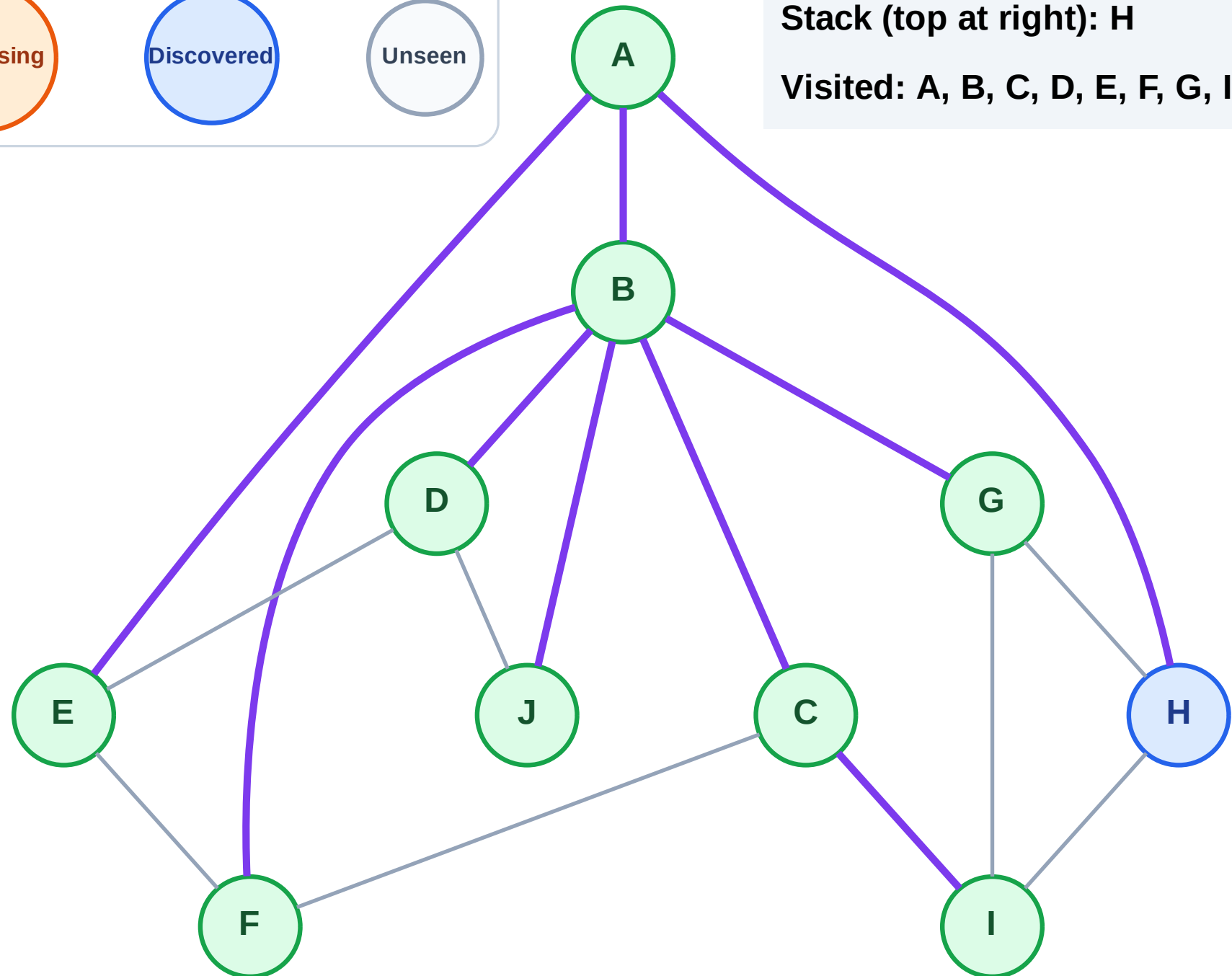
Step 19: No new neighbors from E

Legend



Stack (top at right): H

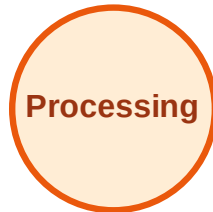
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

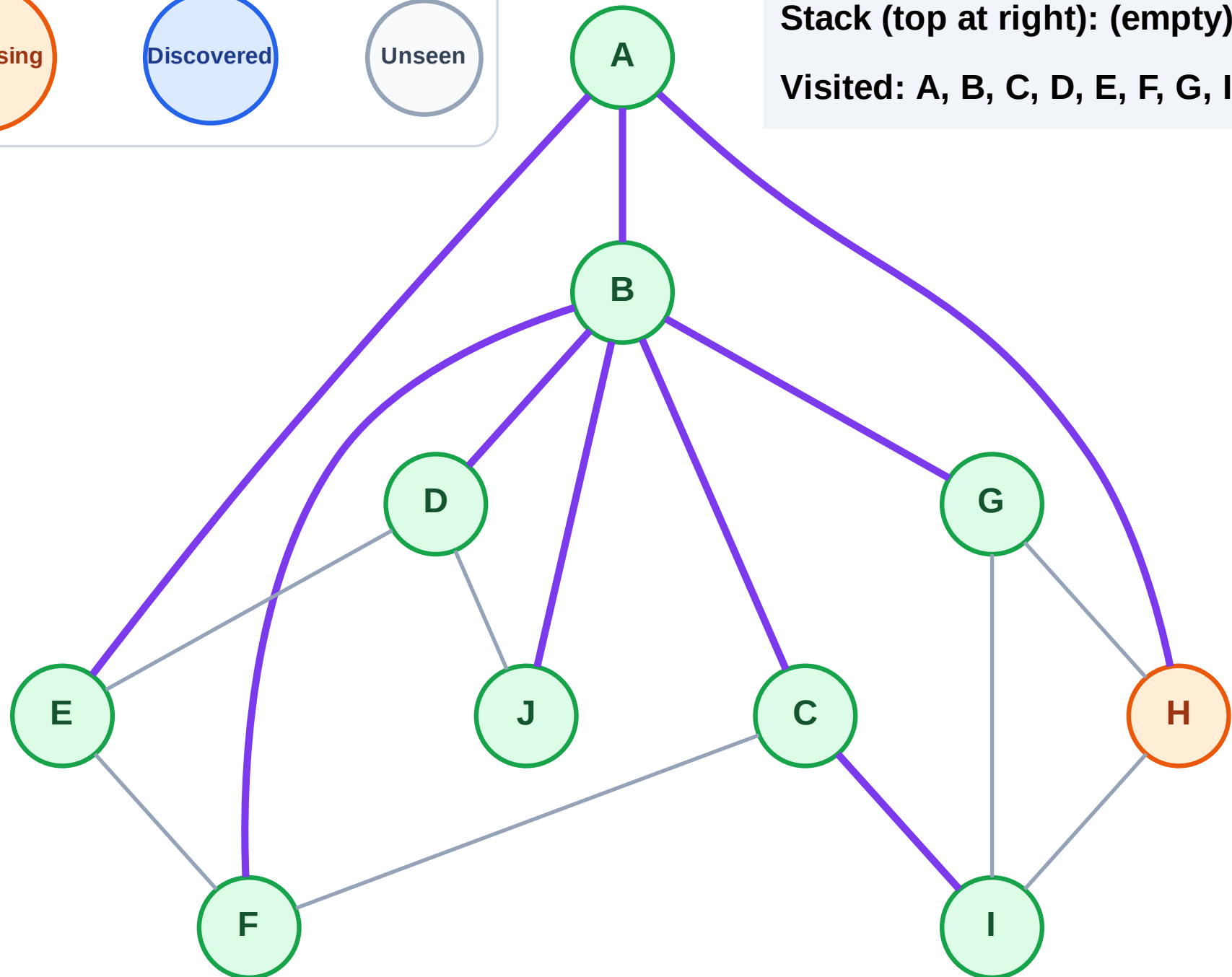
Step 20: Process node H

Legend



Stack (top at right): (empty)

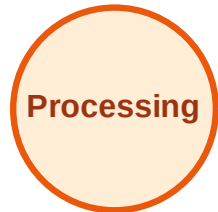
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

